Physics-Aware Multi-Objective Scheduling for an Agile SAR Satellite: Balancing Profit, NESZ, and Geometric Resolution

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Abstract

In agile SAR scheduling, observation timing determines imaging quality through radar-specific geometric and radiometric effects. We present a physics-aware multi-objective formulation of single-satellite Stripmap scheduling that optimizes coverage profit, geometric resolution, and noise-equivalent sigma-zero (NESZ) radiometric quality simultaneously within a non-dominated sorting genetic algorithm III (NSGA-III) framework, and we characterize how the value of physical quality objectives depends on target density. Across 200 scenarios spanning densities 20–500, the benefit of multi-objective optimization is scale-dependent: at sparse density, re-optimizing observation times yields a +34% gain in geometric quality at a 14% coverage cost, and the three-objective variant attains radiometric quality +40% above the greedy-coverage baseline at a 15% coverage cost; under a matched knee-point rule the third objective contributes a controlled +4.9% radiometric margin over the two-objective variant, and this margin only materializes when the selected knee point is radiometric-aware. A single-point squint-minimized heuristic reaches comparable radiometric quality only at roughly half the coverage of the greedy baseline, so the multi-objective advantage is the small coverage price paid for the quality gain; at high densities ($N \geq 300$), the multi-objective solvers converge to the greedy coverage anchor and marginal quality gains fall below 2%. This convergence is conditional on greedy hot-start initialization. The third (radiometric) objective remains density-dependent

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rather than redundant: the geometric–radiometric trade-off is mild in raw visibility geometry ($r \approx -0.36$) but is actively exploited under sparse-density scheduling ($r \approx -0.63$, matching the independent-angle null) and collapses under coverage pressure at high density ($r \approx -0.15$). These results establish an empirical, conditional saturation regime that delimits when physics-aware multi-objective scheduling benefits single-satellite SAR operations.

Keywords: agile SAR, satellite scheduling, multi-objective optimization, NSGA-III, noise-equivalent sigma zero (NESZ), geometric resolution

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1. Introduction

Agile Earth-observation satellites equipped with Synthetic Aperture Radar (SAR) are increasingly central to time-critical monitoring missions, from disaster response to maritime surveillance. Their agility (the ability to slew the antenna beam away from the nadir track) enables a single satellite to image far more targets per orbit than a fixed-pointing platform (Curlander and McDonough, 1991). This flexibility, however, transforms scheduling from a simple feasibility check into a combinatorial optimization problem: for N targets each with a finite visibility window, the scheduler must select a subset and assign observation times that satisfy attitude transition constraints while maximizing some measure of mission value. Throughout this paper, "single-pass" denotes a single satellite operating over a two-orbit scheduling horizon ($\sim$ 200 min, at most one observation per target), not a single overflight. In conventional satellite scheduling, observation quality is commonly represented as a fixed attribute of each pre-computed opportunity rather than a quantity that varies with the selected acquisition time. The scheduling problem therefore reduces to maximizing the number or priority-weighted sum of observed targets subject to temporal feasibility. This assumption underpins most existing satellite scheduling work, from mixed-integer programming formulations to deep reinforcement learning approaches (Chen et al., 2025).

SAR imaging departs fundamentally from this assumption when the scheduler can choose the acquisition time, because image quality depends on the radar geometry at the moment of acquisition: the ground-range resolution degrades with incidence angle ($\rho_{\mathrm{rg}} \propto 1/\sin\theta$), the azimuth resolution broadens with squint angle ($\rho_{\mathrm{az}} \propto 1/\cos\psi_{\mathrm{sq}}$), and the radiometric sensitivity (Noise-Equivalent Sigma Zero, NESZ) deteriorates with slant range as R^3 (Cumming and Wong, 2005). When the scheduler can choose *when* within a visibility window to observe each target (the defining feature of agile satellites), it simultaneously chooses the imaging quality at which each target is acquired. Despite this tight coupling between scheduling and imaging physics, with few exceptions (Kramer and Miller, 2025), existing agile SAR scheduling methods either reduce SAR to a coverage-only objective or apply post hoc quality checks on schedules produced by coverage-maximizing solvers. Few approaches treat physical quality dimensions as primary optimization objectives in the SAR domain.

The problem is NP-hard by reduction from the Orienteering Problem with Time-Dependent Profits (Peng et al., 2019), precluding polynomial-time exact algorithms for all but the smallest instances. The SAR literature has well-established models for NESZ and geometric resolution as functions of observation geometry (Curlander and McDonough, 1991; Cumming and Wong, 2005), and the multi-objective optimization literature provides mature frameworks such as the Non-dominated Sorting Genetic Algorithm III (NSGA-III) (Deb and Jain, 2014) for handling three or more objectives simultaneously. Yet direct integration of these three elements (SAR physical modeling, multi-objective evolutionary algorithms (MOEAs), and satellite scheduling) remains limited. We therefore formulate SAR scheduling with separate, physics-derived geometric and radiometric quality objectives within a MOEA framework.

We formulate a three-objective agile SAR scheduling problem in which the scheduler optimizes (1) total coverage profit, (2) mean geometric image quality (ground-range and azimuth resolution), and (3) mean NESZ radiometric quality simultaneously (denoted f_1^*, f_2, and f_3 below, § 3.2). We compare five solver configurations spanning the methodological spectrum: two greedy heuristics (one profit-only baseline and one squint-minimized variant), a single-objective genetic algorithm with greedy hot-start (GA-P-BL, a control that returns the greedy schedule whenever it fails to improve profit, so it is never worse than G-BL and is numerically identical to G-BL in 199 of 200 scenarios), and two NSGA-III variants with two and three objectives. The solvers are evaluated on 200 systematically varied scenarios across four density classes ($N = 20$ to $N = 500$). Our results show a scale-dependent saturation, not a uniform solver advantage. In sparse scenarios, multi-objective optimization can extract substantial radiometric gains (MOEA-3 raises f_3 by +40% relative to G-BL at a 15% coverage cost) alongside modest geometric gains (MOEA-2, +34% in f_2 at a 14% coverage cost; the best-coverage front point gives +12.5% in f_2). Under a matched radiometric-aware knee rule, the third objective accounts for a controlled +4.9% of this f_3 margin; the margin does not appear under a radiometric-blind selection rule, so it is conditional on the knee-point choice. The single-point squint-minimized heuristic (G-SM) attains slightly higher f_3 than MOEA-3 in sparse scenarios but only at roughly half the coverage of G-BL ($f_1^* \approx 0.51$), so the multi-objective advantage is the small coverage price paid for the quality gain. In dense scenarios, the multi-objective advantage diminishes to negligible levels: greedy methods attain the tested family's coverage reference, and the squint-minimized heuristic offers competitive radiometric quality. This convergence is conditional on the G-BL hot start (§ 6.6), and the fronts are best-known NSGA-III approximations rather than verified Pareto-optimal sets. § 7.1 provides a detailed synthesis of these findings. The specific contributions of the paper are:

1. The discovery and analytical characterization of an empirical saturation regime for greedy-baseline-seeded (G-BL-seeded) multi-objective SAR quality optimization in the single-pass, single-satellite Stripmap setting under the tested configuration: solver convergence to lower-squint geometries attenuates the envelope-level conflict between geometric and radiometric quality while coverage pressure removes the room to trade objectives against each other (§ 6.4), not a physical identity between objectives, making the three-objective formulation indistinguishable in class means from the two-objective one at high densities under the tested search (a trace per-scenario radiometric advantage persists even at the densest class);
2. A controlled ablation study (four objective-function variants over six scenario classes—four density classes plus two targeted robustness classes—totaling 300 scenarios per variant) and initialization controls (§ 6.6) showing that solver outcomes become invariant to the choice of physical objectives at high target densities, that coverage convergence is conditional on greedy-baseline (G-BL) hot-start initialization, and that the fronts are best-known NSGA-III approximations, not verified Pareto-optimal sets;
3. An anchor-independent geometric-coupling baseline: the f_2–f_3 trade-off is mild within raw visibility geometry ($r \approx -0.36$, versus an independent-angle null of $r \approx -0.60$), is restored to the null strength in sparse schedules (-0.63) by squint-aware steering, and collapses to -0.2 to -0.15 at high density under coverage pressure, isolating where the third objective's value is lost to convergence rather than to a mathematical

redundancy;

4. The decomposition of SAR image quality into closed-form NESZ ($\propto R^3$) and geometric quality index ($\sin\theta\cos\psi_{\text{sq}}$, the dimensionless inverse of the ground pixel area since $\rho_{\text{rg}} \propto 1/\sin\theta$ and $\rho_{\text{az}} \propto 1/\cos\psi_{\text{sq}}$) objectives within an NSGA-III framework, distinct from the scalar visibility composite of Kramer and Miller (2025) and the generic image-quality composite of Wei et al. (2025), using mean per-task quality metrics that decouple quality from the number of scheduled tasks. This formulation enables the systematic characterization of quality–coverage trade-offs across two greedy heuristics, a single-objective GA control, and two NSGA-III variants on four scale classes (200 scenarios).

2. Related Work

We examine three lines of research that our study connects: satellite scheduling (2.1), SAR physical quality modeling (2.2), and multi-objective evolutionary optimization (2.3).

2.1. Satellite Scheduling

Satellite scheduling has been studied extensively in the Earth-observation domain. The survey by Ferrari et al. (2025) catalogs over 200 publications spanning exact methods, heuristics, metaheuristics, and learning-based approaches. Exact formulations based on mixed-integer linear programming (MILP) achieve optimality on small instances but fail to scale to operational problem sizes, motivating a broad family of heuristic and metaheuristic alternatives including genetic algorithms, simulated annealing, and tabu search (Wang et al., 2019; Chen et al., 2019). The earliest formal treatments of agile satellite scheduling by Lemaître et al. (2002) and Bianchessi et al. (2007) established the foundational problem structure of time-dependent transition times between observations. Recent work further generalizes these methods to agile satellites, where the freedom to choose observation start times within each visibility window is exploited to pack more observations into a single orbit (Peng et al., 2019; Chen et al., 2024). A parallel line of research has pursued learning-based and hybrid approaches for satellite scheduling. Liu et al. (2024) and Zhou et al. (2025) apply deep reinforcement learning to agile optical satellite scheduling, while Jacquet et al. (2024) combine graph neural networks with Monte Carlo tree search for the same domain. Wang et al. (2026) propose a Lagrangian relaxation-based heuristic for multi-satellite mission planning. Caleiras et al. (2026) formulate super-agile satellite scheduling as a constraint programming problem, capturing the time-dependent transition constraints characteristic of super-agile platforms. Herrmann and Schaub (2023) apply RL to on-board AEOS scheduling, maximizing target collection reward without imaging quality as an objective. Wu et al. (2026) address high-target-density AEOS scheduling with an improved whale optimization algorithm, reporting convergence challenges that echo the density-driven constraint tightening we analyze in § 6.3. Despite their methodological diversity, most prior works treat quality as a fixed property determined at observation-window generation. In contrast, SAR imaging quality varies continuously with acquisition geometry and is directly optimizable within the scheduling window. Chang et al. (2022) propose early MOEAs for SAR satellite scheduling that combine an adaptive large neighborhood search with NSGA-II and NSGA-III selection; their three-objective model (ground-target loss rate, invalid observations, and sensor-activation count) targets operational metrics — the closest prior work to ours — but it omits imaging quality dimensions. Wei et al. (2025) introduce a neural-policy-guided MOEA that incorporates a composite image quality score as an optimization objective. Kramer and Miller (2025) schedule SAR constellation observations with a hybrid metaheuristic using a weighted visibility–performance composite, embedding a SAR-quality proxy in the scheduler, and Yao et al. (2024) develop a bilevel evolutionary algorithm for multi-satellite scheduling that separates task allocation from detailed observation planning. Chang and Zhou (2025) formulate agile Earth-observation scheduling with a bi-objective image-quality-loss/energy model in which quality enters as an attitude-angle proxy attached to each observation opportunity, and Wu et al. (2019) apply NSGA-III to satellite-formation imaging scheduling with coverage- and timeliness-type objectives; in this literature imaging quality is a scalar attribute or coarse proxy rather than a decomposed physical objective. The common limitation of this prior work is that SAR physical quality metrics—noise-equivalent sigma zero (NESZ), azimuth resolution, ground-range resolution—are either absent or approximated by coarse proxies (e.g., penalizing extreme off-nadir angles). To our knowledge, no existing SAR scheduling method treats continuous, physics-derived quality functions as decomposed optimization objectives in their own right: the closest approaches embed aggregated quality proxies rather than separate geometric and radiometric objectives.

2.2. SAR Physical Quality Modeling

The SAR remote sensing literature provides well-established analytical models linking observation geometry to image quality. Curlander and McDonough (1991), Cumming and Wong (2005), and Moreira et al. (2026) give detailed treatments of SAR physics, including the two quality dimensions relevant to scheduling: geometric resolution and radiometric sensitivity. Geometric resolution decomposes into ground-range resolution $\rho_{\text{rg}} = c/(2B \sin\theta)$, where c is the speed of light, B the radar bandwidth, and θ the incidence angle, and azimuth resolution $\rho_{\text{az}} = D/(2\cos\psi_{\text{sq}})$, where D is the antenna length and ψ_{sq} the squint angle. In operational SAR, amplitude weighting widens ρ_{az} by 10–20% (Curlander and McDonough, 1991), but since this is platform-constant, the unweighted form preserves the relative ordering for optimization. The two resolution components respond oppositely to departure from the zero-Doppler geometry ($\psi_{\text{sq}} = 0$, θ minimal): ρ_{az} worsens as $|\psi_{\text{sq}}|$ grows, whereas ρ_{rg} is coarsest at the minimal incidence of closest approach and improves as θ rises away from it—the origin of the θ-axis trade-off formalized in § 3.2. Radiometric sensitivity, measured by the Noise-Equivalent Sigma Zero (NESZ), deteriorates with the cube of slant range: NESZ $\propto R^3$ for Stripmap SAR after azimuth compression gain (Curlander and McDonough, 1991; Moreira et al., 2026). The specific range decomposition and ordering-index approximation used in this work are derived in § 3.2. These quality models are routinely used in SAR system design but their use in scheduling remains limited to post hoc validation of fixed observation plans (Kramer and Miller, 2024) or binary feasibility filtering at the observation-window level (Li et al., 2026). The scheduling literature and the SAR quality modeling literature have developed in parallel: schedulers treat quality as a constraint, while SAR engineers compute quality after the schedule is fixed. Our study connects these two domains by embedding continuous quality functions directly into the optimization objective space.

2.3. Multi-Objective Evolutionary Optimization

Multi-objective evolutionary algorithms (MOEAs) are a standard toolkit for problems with competing objectives. The NSGA-III algorithm (Deb and Jain, 2014) extends the NSGA-II framework (Deb et al., 2002) to many-objective problems through reference-point-based diversity preservation. The hypervolume (HV) indicator is widely used for comparing Pareto fronts (Zitzler and Thiele, 1999), but its sensitivity to frontier cardinality creates interpretational challenges when mixing single-objective and multi-objective solvers, a methodological issue we address in § 6.4. To our knowledge, MOEAs have not been applied to SAR scheduling with physical quality objectives; the closest work uses single-objective optimization with post hoc quality evaluation. Our NSGA-III formulation makes these trade-offs explicit by placing coverage profit, geometric quality, and radiometric quality in a unified multi-objective framework.

3. Problem Formulation

3.1. System Model

We consider a single agile SAR satellite in a circular orbit at altitude H, operating in Stripmap mode, which provides the widest continuous swath with uniform quality and analytically tractable NESZ and resolution expressions (Curlander and McDonough, 1991). A problem instance $\mathcal{I} = (\mathcal{T}, \mathcal{P}, \mathcal{O})$ consists of a set of N ground targets $\mathcal{T}$, a satellite platform $\mathcal{P}$, and a SAR imaging model $\mathcal{O}$.

Task Set.. Each task $i \in \{1, \dots, N\}$ is specified by its geographic coordinates $\mathbf{r}_i = (\phi_i, \lambda_i)$, a priority weight $p_i \in \mathbb{R}^+$, a visibility window $\mathcal{W}_i = [a_i, b_i]$ during which the satellite can observe the target, and a fixed observation duration d_i.

Satellite Platform.. The satellite is characterized by its maximum slew rate $\omega_{\max}$ (a conservative single-axis bound on three-axis attitude maneuvers), settling time τ_{settle} after each maneuver, per-orbit energy budget $E_{\max}$ and memory budget $M_{\max}$, and per-observation energy e_i and memory m_i consumption.

SAR Observables.. Given an observation time t_i, the satellite-to-target line-of-sight (LOS) vector $\mathbf{l}(t_i)$ in the Local-Vertical Local-Horizontal (LVLH) frame is fully determined by the satellite's orbital state. From $\mathbf{l}(t_i)$ we derive four geometric quantities deterministically: slant range $R = |\mathbf{l}|$, off-nadir angle $\xi = \arctan 2(\sqrt{l_x^2 + l_y^2},\, l_z)$, incidence angle θ (via the Earth-curvature-corrected mapping $\theta = \arcsin\left(\frac{R_e+H}{R_e} \sin\xi\right)$ where R_e is Earth's mean radius), and squint angle $\psi_{\text{sq}} = \arcsin(|l_x|/R)$ (LVLH x-axis along-track, z-axis nadir-pointing), defined as the angular deviation of the LOS vector from the zero-Doppler plane (Curlander and McDonough, 1991). For the objectives we also use the elevation-plane incidence angle θ_{elev}, defined through the identity $\cos\theta_{\text{elev}} = \cos\xi / \cos\psi_{\text{sq}}$ (i.e., the incidence angle of the LOS projection onto the cross-track plane); this isolates the cross-track ground-range geometry from the along-track squint contribution. θ_{elev} is the satellite-centered elevation-plane angle; it differs from the target-side incidence angle of §3.1 by a few degrees due to Earth curvature at these orbital altitudes, but it preserves the cross-track ordering of candidate views, so using it in the geometric-quality objective of § 3.2 is ordering-consistent. A critical property is that *all* quality-relevant geometry is a deterministic function of the single decision t_i.

Decision Variables.. A schedule consists of a task selection vector $\mathbf{x} \in \{0,1\}^N$, an ordered sequence $S = (s_1, \ldots, s_m)$ of the m selected tasks, and a vector of observation start times $\mathbf{t} = (t_{s_1}, \ldots, t_{s_m})$. All geometric parameters (θ_i, $\psi_{\text{sq},i}$, R_i) are *derived* from t_i rather than independent decision variables.

3.2. Objective Functions

We formulate a three-objective optimization problem that captures the tension between scheduling throughput and SAR-specific imaging quality. All quality objectives use the *mean* over scheduled tasks, making them independent of the number of scheduled tasks $m = |\mathcal{S}|$ and thereby producing a genuine "quantity versus quality" Pareto trade-off.

Objective 1: Coverage Profit.. The normalized coverage profit (Eq. 1) is defined relative to the greedy baseline

$$f_1^* = \frac{1}{f_1^{\text{G-BL}}} \sum_{i \in \mathcal{S}} p_i \tag{1}$$

where $f_1^{\text{G-BL}}$ is the profit achieved by the profit-only greedy scheduler (G-BL, § 4.2) on the same scenario. Normalization by G-BL rather than GA-P-BL is conservative for the MOEA coverage deficit: GA-P-BL output is identical to G-BL in all but one of the 200 scenarios ($f_1^* = 1.00$ at every scale, Table 3), so G-BL-normalized f_1^* values equal the coverage comparison to the single-objective GA.

Objective 2: Geometric Image Quality.. The ground pixel area of a SAR image is the product of ground-range resolution $\rho_{\text{rg}} = c/(2B\sin\theta)$ and azimuth resolution $\rho_{\text{az}} = D/(2\cos\psi_{\text{sq}})$. We define a dimensionless geometric quality index:

$$f_2 = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} \sin\theta_{\text{elev},i} \cdot \cos\psi_{\text{sq},i} \tag{2}$$

Higher f_2 indicates smaller ground pixel area (better geometric resolution). The quality indices are unweighted means over the scheduled set, orthogonal to the priority weights used in f_1: they reflect the average imaging geometry of the selected tasks rather than biasing quality toward high-priority targets.

Objective 3: NESZ Radiometric Quality.. Noise-Equivalent Sigma Zero (NESZ) measures the minimum detectable backscatter. For Stripmap SAR, NESZ degrades with the cube of slant range, NESZ $\propto R^3$, after accounting for azimuth compression gain (Curlander and McDonough, 1991). Using the full off-nadir angle ξ for the R^3 factor (which already contains the squint contribution through $\cos\xi = \cos\theta_{\text{elev}}\cos\psi_{\text{sq}}$, so no separate squint factor is needed), we define the radiometric quality index as:

$$f_3 = \frac{1}{|\mathcal{S}|}\sum_{i\in\mathcal{S}} \cos^3 \xi_i \tag{3}$$

Higher f_3 indicates lower NESZ (better radiometric sensitivity).

Modeling choices behind Eq. 3.. Equation 3 is grounded in a 3D geometric approximation that decomposes the slant range as $R \approx R_{\text{elev}}/\cos\psi_{\text{sq}}$ (where R_{elev} is the elevation-plane range at zero squint), yielding the equivalent form NESZ $\propto 1/(\cos^3\theta_{\text{elev}} \cdot \cos^3\psi_{\text{sq}})$. This plane-Earth range model omits the Earth's curvature and the antenna elevation-pattern roll-off; at the 47° edge of the tested incidence range it overestimates slant range by roughly 5% and R^3 by roughly 17%, and it suppresses the $\sin\theta$ factor that a ground-resolution reference area adds to standard NESZ expressions. It is therefore used here as a relative ordering index across observation geometries, not as an absolute radiometric prediction – the comparisons in §5.3 and below depend only on the ordering it induces. This ordering defense is strongest for within-window timing selection, where incidence varies by only a few degrees (about 1.5° mean within-window range in the S1 scenarios; §6.5); across targets or windows that span the full 18°–47° incidence range, the suppressed $\sin\theta$ factor varies by roughly 2.4×, so f_3 should be read as a geometry proxy rather than an absolute NESZ ranking across widely separated incidence angles. In the schedules produced by the solvers below, the per-task squint angle is moderate rather than near-zero (the mean $|\psi_{\text{sq}}|$ across the four density classes is 16–24°, with a 90th percentile of 32–39°), so the $\cos^3\psi_{\text{sq}}$ factor contributes on the order of 10–25% of the f_3 range rather than the few percent it would at near-zero squint. The proportionality NESZ $\propto R^3$ captures the dominant geometric dependence; platform-constant terms (noise figure, transmit power, processing gain) do not affect the relative ordering of observation geometries. This cubic dependence means even modest deviations from the optimal viewing geometry make fine observation timing critical for radiometric performance.

Physical Trade-Off.. The two quality objectives partition SAR imaging performance by physical mechanism: f_2 captures spatial resolution (geometric), f_3 captures signal-to-noise ratio (radiometric). Their optima are not coincident; f_3 peaks at the zero-Doppler point (ξ minimal, $\psi_{\mathrm{sq}} = 0$), whereas f_2 benefits from moderately higher incidence because $\sin\theta_{\mathrm{elev}}$ grows with the elevation-plane incidence angle. This creates a genuine trade-off: pursuing the cleanest NESZ (zero-Doppler observation) forces acceptance of stretched ground-range pixels; pursuing optimal ground pixel area requires deviating from the zero-Doppler point and accepting elevated radiometric noise. The squint angle ψ_{sq} appears in both f_2 and f_3 (Eqs. 2–3): reducing ψ_{sq} simultaneously improves both resolution components through the $\cos\psi_{\mathrm{sq}}$ factor (azimuth resolution enters f_2 directly, and $\cos\xi = \cos\theta_{\mathrm{elev}}\cos\psi_{\mathrm{sq}}$ carries the same factor into the R^3 radiometric index), so squint is an alignment rather than a trade-off axis. The residual two-axis structure is $[\theta_{\mathrm{elev}}, \psi_{\mathrm{sq}}]$: the θ_{elev} axis carries the genuine conflict (f_2 grows with $\sin\theta_{\mathrm{elev}}$, f_3 with $\cos^3\theta_{\mathrm{elev}}$ at fixed squint), while ψ_{sq} enters both objectives with the same sign. In practice, the zero-squint point coincides with the time of closest approach, which minimizes the slant range and, for a spherical Earth, approximately minimizes the off-nadir angle ξ (i.e. maximizes $\cos\xi$ and hence f_3); solver convergence to low-squint regimes removes the alignment axis and leaves the θ_{elev} trade-off, whose exploitation is what saturates under coverage pressure at high density (Eq. 6 in § 6.4 and its axis decomposition for the separation into definitional and scheduler-induced components).

Because θ and ψ_{sq} are derived from the same LOS vector (§ 3.1), the per-task f_2 and f_3 contributions share geometric structure; its quantitative characterization is presented in § 6.4.

3.3. Constraints

Observation-window membership is treated as the domain of each start-time variable rather than as a separate numbered constraint:

$$t_i \in [a_i, b_i - d_i] \quad \forall i \in \mathcal{S}. \tag{4}$$

The remaining four constraints couple task selection, acquisition geometry, temporal feasibility, and resource use. The incidence-angle and squint-angle constraints (C1) are enforced during visibility-window generation; C2–C4 are checked inline by every solver.

C1: Incidence and Squint Angle Constraints.. Each selected acquisition must satisfy the platform incidence-angle range, $\theta^{\min} \leq |\theta_i| \leq \theta^{\max}$, and the squint-angle limit $|\psi_{\mathrm{sq},i}| \leq \psi_{\mathrm{sq}}^{\max}$, where $\psi_{\mathrm{sq}}^{\max}$ is the satellite's maximum achievable squint; we adopt $|\psi_{\mathrm{sq}}| \leq 45^\circ$ as a controlled experimental agility envelope. This bound is an experimental setting, not a claim about any specific platform's demonstrated operating mode or capability. Within C1-feasible visibility geometry the usable envelope is effectively narrower: window generation filters at the 10 s sampling grid, and the θ–ψ_{sq} coupling of the LOS vector (the window-level $|\psi_{\mathrm{sq}}|$ 95th percentile is 37° and the maximum 45°; §6.4) confines feasible acquisitions; the selected schedules concentrate at mean $|\psi_{\mathrm{sq}}|$ of 16–24°. Both are observation-geometry limits applied before scheduling.

Physical modeling scope.. The quality objectives capture first-order geometric and radiometric effects (ground-range resolution, azimuth broadening, R^3 NESZ scaling). Second-order radar effects are treated as platform-level constants and are not modeled: the Doppler centroid $f_{\mathrm{dc}} \approx (2v/\lambda)\sin\psi_{\mathrm{sq}}$ (of order 10^2 kHz at the scheduled 16–24° squints, versus a PRF of order 10^0–10^1 kHz, and hence routinely accommodated by yaw steering or Doppler-centroid tracking in operational Stripmap systems), the azimuth antenna gain roll-off with squint, and squint-dependent azimuth processing. This idealization preserves the relative ordering of acquisition geometries *within* the simulated platform family, which is what the solver comparisons act on; absolute performance claims for platforms without Doppler compensation at large squint are out of scope. The sensitivity of the recommendations to the agility envelope itself is tested directly in §7.2.

C2: Attitude Maneuver and Non-Overlap.. Between consecutive observations, the satellite must rotate through the angular separation of their LOS vectors. Let $\mathbf{l}_k$ and $\mathbf{l}_{k+1}$ be the LVLH-frame LOS vectors at t_{s_k} and $t_{s_{k+1}}$. The angular gap $\Delta\eta_k = \arccos(\mathbf{l}_k \cdot \mathbf{l}_{k+1}/(|\mathbf{l}_k||\mathbf{l}_{k+1}|))$ must be traversed within the available inter-observation time:

$$t_{s_{k+1}} \geq t_{s_k} + d_{s_k} + \frac{\Delta\eta_k}{\omega_{\max}} + \tau_{\mathrm{settle}} \quad \forall k \in \{1, \ldots, m-1\}. \tag{5}$$

This inequality also enforces non-overlapping task-execution intervals. It is the core coupling in our formulation: changing t_{s_k} shifts $\mathbf{l}_k$, affecting both imaging quality (via θ_k and $\psi_{\mathrm{sq},k}$) and transition feasibility (via $\Delta\eta_k$).

The tightness of C2 scales with task density: as N grows, the fraction of task pairs whose required angular transition $\Delta\eta/\omega_{\max}$ exceeds the available inter-observation time increases, tightening the constraint and making C2 the dominant active constraint in dense regimes.

C3–C4: Energy and Memory Budgets.. $\sum_{i\in\mathcal{S}} e_i \le E_{\max}$ and $\sum_{i\in\mathcal{S}} m_i \le M_{\max}$, where $E_{\max} = 10^7$ and $M_{\max} = 10^{11}$ (arbitrary units) are set sufficiently high to be non-binding under all tested scenarios (typical per-task $e_i = 5 \times 10^4$, $m_i = 5 \times 10^8$, far below the caps); they are retained for completeness.

4. Methodology

4.1. Solver Overview

Table 1 summarizes the five configurations compared in this study. We describe the solvers in order of increasing sophistication: from greedy heuristics (G-BL, G-SM) through single-objective metaheuristic (GA-P-BL), to multi-objective evolutionary algorithms (MOEA-2, MOEA-3).

Table 1: Solvers: greedy baselines (G-BL, G-SM), single-objective GA with hot-start (GA-P-BL), and NSGA-III variants with two and three objectives (MOEA-2, MOEA-3)

Label	**Type**	**Objectives**	**Hot-Start**
G-BL	Greedy heuristic	max f_1 (profit-only)	n/a
G-SM	Greedy heuristic	max f_1 + squint minimization	n/a
GA-P-BL	Single-objective GA	max f_1 (G-BL seeded)	std = 0.5
MOEA-2	NSGA-III (2-objective)	$f_1^* + f_2$ (geometric quality)	std = 0.5
MOEA-3	NSGA-III (3-objective)	$f_1^* + f_2 + f_3$ (geo.+NESZ)	std = 0.5

Single-objective solvers (G-BL, G-SM, GA-P-BL) optimize only f_1; their f_2 and f_3 values are computed post hoc on the final schedule for comparison with the multi-objective solvers. The five configurations differ primarily in objective-function configuration rather than algorithmic paradigm. This design is intentional: it isolates the effect of adding physical quality objectives, which is the paper's research question. A comparison across algorithmic families (e.g., NSGA-III vs. MOEA/D (decomposition-based) vs. SMS-EMOA (hypervolume-based)) is left to future work (§ 7.4).

4.2. G-BL: Greedy Baseline

G-BL represents the conventional scheduling approach: it makes no use of SAR imaging physics when ordering or selecting tasks. Tasks are sorted by earliest start time t_{start} and greedily inserted into the schedule at their earliest feasible time. This earliest-time placement fixes each task's observation geometry implicitly through the window's incidence–squint profile rather than through any quality criterion. If the attitude transition constraint (C2) is violated for a candidate task, its start time is delayed within the window until the constraint is satisfied; if no feasible delay exists, the task is dropped. Since G-BL selects observation times solely for feasibility, its f_2 and f_3 values are incidental by-products of the window geometry. We compute these post hoc on the final schedule to serve as the baseline against which physics-aware solvers are compared.

4.3. G-SM: Greedy Squint-Minimized

G-SM shares G-BL's greedy structure (sort by earliest start time) but modifies the observation time selection: within each visibility window, a task is placed at the discretized time step that minimizes squint angle $\psi_{\text{sq}} = \arcsin(|l_x|/R)$. G-SM minimizes squint angle as a computationally cheap proxy for maximizing f_3 (NESZ radiometric quality); the correlation between squint minimization and f_3 improvement is confirmed by the large effect size (Cliff's $\delta = +0.99$, § 6.2). This strategy serves as a physics-aware greedy baseline: it answers the question "what quality improvement can a simple squint-minimization heuristic achieve, without any multi-objective search?" The squint-minimizing time selection incurs a profit penalty because the optimal-squint time may lie later in the window, reducing the remaining time for subsequent tasks (constraint C2). As with G-BL, f_2 and f_3 are post hoc outputs.

4.4. GA-P-BL: Single-Objective GA with Hot-Start

GA-P-BL tests whether a genetic algorithm optimizing only f_1 can outperform greedy scheduling by exploiting temporal flexibility. Each chromosome encodes a vector of normalized start times $\boldsymbol{\tau} \in [0,1]^N$, where task i's actual start time is $t_i = t_i^{\text{earliest}} + \tau_i \cdot (t_i^{\text{latest}} - d_i - t_i^{\text{earliest}})$, alongside a binary selection vector $\mathbf{x} \in \{0,1\}^N$; both are decision variables (a $2N$-dimensional chromosome). The population of 100 individuals is initialized from G-BL's solution with Gaussian noise (std $= 0.5$) added to the full $2N$-dimensional chromosome. The GA runs for 200 generations with simulated binary crossover ($p_c = 0.9$, $\eta_c = 20$) and polynomial mutation ($p_m = 0.1$, $\eta_m = 20$). f_2 and f_3 are computed post hoc on the best-found schedule. After optimization, a never-worse-than-baseline guard compares the best GA schedule with G-BL: if the GA's f_1 falls short, the greedy schedule is returned instead, guaranteeing $f_1^{\text{GA-P-BL}} \geq f_1^{\text{G-BL}}$ by construction. In practice this guard makes GA-P-BL output numerically identical to G-BL in 199 of 200 scenarios (the single exception improves profit by 13.5%; mean f_1^* differs by less than 0.001) (§6.1); GA-P-BL therefore acts as a control that confirms the greedy profit baseline rather than as an independent solver.

4.5. MOEA-2 and MOEA-3: Multi-Objective Evolutionary Algorithms

MOEA-2 and MOEA-3 extend the GA-P-BL chromosome encoding to multi-objective optimization via the NSGA-III framework (Deb and Jain, 2014). Both use a population of 100 individuals over 200 generations with the same genetic operators as GA-P-BL. Reference points are generated via the Das–Dennis systematic approach (Das and Dennis, 1998), producing $H = \binom{n_{\text{obj}}+p-1}{p}$ structured reference directions, where n_{obj} is the number of objectives and $p = 12$ the divisions per objective dimension ($H = 13$ for MOEA-2, $H = 91$ for MOEA-3). The entire population is hot-started from the G-BL solution perturbed by Gaussian noise (std $= 0.5$), and an at-least-one-task constraint prevents empty solutions from dominating the Pareto front. As every solver is hot-started from the non-empty G-BL solution (§ 4.2), this constraint is inactive in the reported experiments. MOEA-2 optimizes two objectives: f_1^* (normalized profit) and f_2 (mean geometric quality, Eq. 2). MOEA-3 adds f_3 (mean NESZ radiometric quality, Eq. 3) as a third objective. The comparison between MOEA-2 and MOEA-3 directly tests whether adding the NESZ objective (f_3) produces Pareto-front differentiation beyond what f_2 alone captures, or whether the θ/ψ_{sq} geometric coupling renders the third objective redundant in the single-pass, single-satellite setting.

5. Experimental Setup

5.1. Scenario Design

Table 2 defines the four scenario classes.

Table 2: Scenario classes: four density levels (S1–S4) with $N = 20, 100, 300, 500$ targets, 50 scenarios per class

Class	Targets	Look Direction	Description
S1	20	Both	Small, low-density
S2	100	Both	Medium, moderate density
S3	300	Both	Medium-large, transition region
S4	500	Both	Large, high-density

All scenarios share a common satellite platform abstracted as a generic C-band Stripmap SAR with bilateral-looking capability. The platform follows a circular sun-synchronous orbit at 693 km altitude (inclination $\approx 97.8°$, LTAN 06:00, dawn–dusk) and carries a C-band Stripmap instrument with antenna length $D = 12.3$ m and bandwidth $B = 56.5$ MHz. It accesses both left- and right-side incidence angles in the range $[18°, 47°]$ (signed convention: $\theta \in [-47°, -18°] \cup [18°, 47°]$). This generic bilateral configuration is selected to exercise the full incidence–squint trade-off space.

Targets are distributed along the satellite ground track within the accessible observation band (cross-track half-width roughly ± 608 km, a ground central angle of about $\pm 5.5°$, corresponding to the 47° incidence limit). Within each class, five scenario variants span the target distribution (uniform, clustered, and mixed) and geometry: variants A–C modulate only the target distribution, while variants D and E additionally modulate the observation geometry, with a class-dependent role — S1-D and S3-D use a single orbital revolution (yielding fewer and narrower windows), S2-E places targets at high latitudes ($\pm$40–70°), and S4-D narrows the incidence ceiling from 47° to 25°. The 50 scenarios per class consist of these 5 variants $\times$ 10 random seeds each; the density trend reported in §6.3 is robust to these within-class modulations (e.g., the sparse-regime coverage deficit appears across all S1 variants, and the densest S4 class remains at or near full coverage even for the variants with fewer or tighter windows). The four density classes (S1–S4) span sparse reconnaissance ($N = 20$) to dense monitoring ($N = 500$). Other intermediate densities ($N = 50, 400$) are omitted to limit computational cost. To directly constrain the transition midpoint, two additional intermediate classes—S7 ($N = 150$) and S8 ($N = 200$), 50 scenarios each generated under the same generic C-band bilateral configuration—were run for MOEA-2 and the two greedy baselines (G-BL, G-SM) and are used solely in the scale-sensitivity analysis in § 6.3; they do not enter the five-solver pooled comparisons.

Each task is assigned a priority weight p_i drawn uniformly from the integer set $\{1, 2, \ldots, 10\}$, and a fixed observation duration $d_i = 30\,\text{s}$. Visibility windows are computed via orbit propagation over two orbital periods (∼200 min), yielding 1–2 windows per target on average, each typically spanning about a minute (median 1.0–1.2 min across the density classes). Candidate observation times are sampled at a 10 s step within each window, and all reported numbers correspond to a fixed scenario manifest (no manifest revision postdates the results). Windows are filtered by incidence-angle, squint, and minimum-elevation constraints at generation time; the quality indices are then evaluated at each scheduled acquisition time, and C1 feasibility is additionally verified over the full 30 s observation interval (a 0.5 s-resolution scan of all 76,190 windows across the four density classes reports zero violations; the median change of squint over any 30 s within-window sub-interval is 11°, 95th percentile 15°).

5.2. Solver Configuration

All five configurations run on a single CPU core to ensure a uniform computational environment. For orientation, single-core wall-clock per scenario is milliseconds for the greedy heuristics and a median of about 22 s (S1), 66 s (S2), 220 s (S3), and 260 s (S4) for the NSGA-III variants; the study's claims concern solution quality rather than computational efficiency, so these are order-of-magnitude indications rather than a benchmark. G-BL and G-SM are deterministic greedy heuristics with no tunable hyperparameters. GA-P-BL and the MOEA variants use the population size, generation budget, and genetic operators described in §4.4 and §4.5, with the entire MOEA population hot-started from the G-BL solution. The NSGA-III selection operator follows the reference-point-based selection of Deb and Jain (2014).

5.3. Evaluation Metrics

We treat per-objective comparisons paired with Cliff's δ effect sizes as the primary quality indicator, because single-point greedy solvers and multi-point NSGA-III fronts contribute different numbers of points to the objective space. Hypervolume (HV) is reported as a secondary, cardinality-sensitive diagnostic: as shown in § 6.4, its ranking inverts if the single-point G-SM is included (its HV of 0.2341 exceeds MOEA-3's 0.0893), so HV is not interpreted as a standalone quality judgment. Statistical significance is assessed via the Friedman test for overall differences across the five configurations $\times$ 200 scenarios, followed by pairwise Wilcoxon signed-rank comparisons with Bonferroni correction ($\alpha = 0.05/10 = 0.005$); because G-BL and GA-P-BL coincide in 199 of 200 scenarios (and never differ to GA-P-BL's disadvantage by construction), the Friedman statistic is conservative toward fewer distinct rankings and is driven largely by within-scenario dispersion of the two multi-point NSGA-III solvers. Per-objective and ablation comparisons are reported with raw p-values alongside the Bonferroni threshold. Effect sizes are measured by Cliff's δ, classified following Romano et al. (2006) ($|\delta| < 0.147$ negligible, < 0.33 small, < 0.474 medium, ≥ 0.474 large). For paired within-scenario solver comparisons we report the matched-pairs sign statistic $\delta_{\text{pairs}} = (W - L)/n$, which coincides with Cliff's δ when each scenario is evaluated as a matched pair; the conventional unpaired form $\delta = \big(\#\{x_i > y_j\} - \#\{x_i < y_j\}\big)/(nm)$ gives smaller magnitudes for the same comparisons (e.g. +0.57 versus +0.88 for the sparse MOEA-3/MOEA-2 f_3 comparison), and Romano's thresholds are applied to that conventional form wherever both are quoted. Hypervolume is computed with reference point $(0, 0, 0)$ in the min–max-normalized objective space, where the six min/max bounds (f_1: 0.06–1.16; f_2: 0.11–0.56; f_3: 0.45–0.87) are pooled across all solvers, scenarios, and every persisted non-dominated frontier point (not only knee solutions); per-scenario HV is then averaged within and across classes. This shared normalization framework makes cross-solver HV values directly comparable as relative rankings, subject to the cardinality caveat stated above and detailed in § 6.4 (HV also reflects the cardinality of each solver's reported non-dominated front).

For each MOEA scenario, a single representative solution is selected from the feasible non-dominated front as the knee point: the objectives are min–max normalized across that scenario's front, and the point maximizing the normalized sum ($f_1 + f_2$ for MOEA-2, $f_1 + f_2 + f_3$ for MOEA-3) is reported in Table 3 and the per-objective means. The full front is retained for HV and for the Pareto compression analysis (§ 6.4). We verified the selector sensitivity directly: at $N \geq 100$ the knee and best-f_1 points give class-mean f_2 and f_3 within 0.005, because the dense-regime fronts are compressed (S4 spans only about 0.014 in f_2); at sparse S1 the knee and best-f_1 means differ by about 0.04–0.06 in f_2 and f_3, which is expected and intentional — Table 3 reports the balancing knee rather than either single-objective extreme. The S1 fronts bracket the knee ($f_2 = 0.376$, $f_3 = 0.698$, $f_1^* = 0.85$): the best-f_2 point reaches $f_2 = 0.449$ at $f_1^* = 0.58$, and the best-f_3 point reaches $f_3 = 0.738$ at $f_1^* = 0.68$, so the headline figures are conservative compromise values rather than achievable optima. Quality gains are also class-mean figures: because the objectives are unweighted per-task means, a positive class-mean f_2/f_3 gain does not selectively favor high-priority targets, and per-scenario spread persists even at dense classes (S4 MOEA-3 f_2 10th percentile 0.291 vs. G-BL mean 0.336).

5.4. Experiment Design Overview

This study comprises three classes of experiments. The first is the main experiment, running all five configurations (G-BL, G-SM, the GA-P-BL control, MOEA-2, MOEA-3) on 50 scenarios per density class (S1–S4), totaling 200 scenarios. Results from the main experiment are presented in §6.1–§6.4 and §6.2. The second is the ablation experiment, which successively removes the squint component (variant B), the incidence-angle component (variant C), and all physical objectives (variant D) from the MOEA-3 framework, running each variant on the same six scenario classes (300 scenarios; §6.5). The third comprises robustness control experiments—hot-start dependence, perturbation-sensitivity sweep, random-search calibration, and search-budget control— which verify that the main and ablation findings are not artifacts of initialization or parameter choices (§6.6).

6. Results and Analysis

6.1. Overall Solver Performance

Table 3 reports all five configurations across all four scenario classes. The full-scale view shows why the value of multi-objective optimization cannot be judged from the densest scenario class (S4) alone. At the sparsest density S1, MOEA-2 improves f_2 by +34% relative to G-BL (0.393 vs. 0.295) while accepting a 14% coverage reduction ($f_1^* = 0.86$, knee-point value; the best-coverage front point gives +12.5%), and MOEA-3 raises f_3 by +40% (0.698 vs. 0.497) at a 15% coverage cost ($f_1^* = 0.85$). At S2 the performance gains diminish, and at S3–S4 MOEA coverage and quality converge to the G-BL reference. Across every scale, GA-P-BL output is numerically identical to G-BL in 199 of 200 scenarios (mean f_2 and f_3 within 0.001 of G-BL, and the single differing scenario improves profit), because its never-worse-than-baseline guard (Sec. 4.4) replaces the GA schedule whenever it fails to beat the greedy anchor on the single profit objective. GA-P-BL therefore serves as a control that confirms the greedy coverage baseline rather than an independent solver. G-SM occupies a different operating regime, exchanging coverage and f_2 for substantially higher f_3.

Table 3: Solver performance across all scenario classes (mean ± standard deviation over 50 scenarios per class). Higher values are preferred for f_1^*, f_2, and f_3.

Class	**Solver**	f_1^*	f_2	f_3
S1 ($N = 20$)	G-BL	1.00 ± 0.00	0.295 ± 0.037	0.497 ± 0.044
	G-SM	0.51 ± 0.16	0.389 ± 0.086	0.722 ± 0.058
	GA-P-BL	1.00 ± 0.02	0.295 ± 0.037	0.498 ± 0.044
	MOEA-2	0.86 ± 0.13	0.393 ± 0.055	0.657 ± 0.040
	MOEA-3	0.85 ± 0.12	0.376 ± 0.050	0.698 ± 0.039
S2 ($N = 100$)	G-BL	1.00 ± 0.00	0.317 ± 0.026	0.561 ± 0.060
	G-SM	0.35 ± 0.09	0.357 ± 0.078	0.735 ± 0.055
	GA-P-BL	1.00 ± 0.00	0.317 ± 0.026	0.561 ± 0.060
	MOEA-2	0.98 ± 0.03	0.329 ± 0.030	0.589 ± 0.044
	MOEA-3	0.98 ± 0.04	0.329 ± 0.030	0.597 ± 0.042
S3 ($N = 300$)	G-BL	1.00 ± 0.00	0.342 ± 0.026	0.616 ± 0.042
	G-SM	0.29 ± 0.04	0.317 ± 0.055	0.756 ± 0.036
	GA-P-BL	1.00 ± 0.00	0.342 ± 0.026	0.616 ± 0.042
	MOEA-2	1.00 ± 0.00	0.343 ± 0.026	0.622 ± 0.040
	MOEA-3	1.00 ± 0.01	0.344 ± 0.026	0.623 ± 0.040
S4 ($N = 500$)	G-BL	1.00 ± 0.00	0.336 ± 0.039	0.675 ± 0.074
	G-SM	0.37 ± 0.12	0.291 ± 0.067	0.786 ± 0.046
	GA-P-BL	1.00 ± 0.00	0.336 ± 0.039	0.675 ± 0.074
	MOEA-2	1.00 ± 0.01	0.337 ± 0.038	0.680 ± 0.073
	MOEA-3	1.00 ± 0.01	0.338 ± 0.038	0.680 ± 0.073

Table 4: Hypervolume across 200 scenarios for the two multi-objective solvers, with G-BL as a single-point reference (single-point solvers G-SM and GA-P-BL are omitted from this comparison; see § 6.4). HV comparisons between single-point and multi-point solvers are confounded by point-count asymmetry (see text).

Solver	**HV**	**SD**
MOEA-3	0.0893	0.1053
MOEA-2	0.0668	0.0642
G-BL	0.0234	0.0150

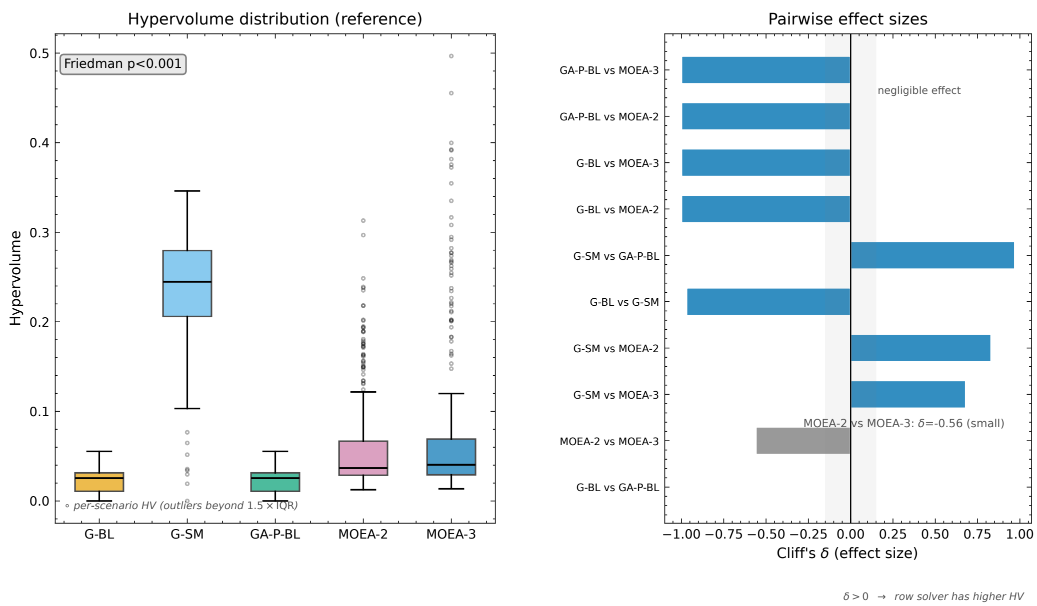

Figure 1: Statistical validation: (left) hypervolume distribution; (right) pairwise effect sizes as the matched-pairs sign statistic $\delta_{\text{pairs}} = (W - L)/n$, which equals Cliff's δ for the within-scenario pairing; the gray band marks the negligible-effect range ($|\delta| < 0.147$ of Romano et al. (2006); the conventional unpaired δ is weaker for the within-pair comparisons, see §5.3).

Fig. 1 summarizes the statistical evidence behind these comparisons. The left panel shows the per-scenario hypervolume (HV) distribution of each solver under a shared min–max normalization: the multi-point NSGA-III solvers attain higher pooled HV than G-BL, while the extreme operating point of G-SM yields an HV comparable to the NSGA-III variants. The right panel reports pairwise effect sizes on HV as the matched-pairs sign statistic; bars outside the gray band ($|\delta| < 0.147$) exceed the negligible-effect threshold, and the direction convention is that positive δ favors the row solver. Because HV rewards front cardinality as well as front quality, these rankings are interpreted only as secondary diagnostics — the point-count confound is analyzed in § 6.4.

The following sections deepen this overall analysis from five perspectives: §6.2 examines the special trade-off of squint-aware scheduling, §6.3 investigates scale sensitivity (how multi-objective advantage varies with target density), §6.4 reveals the geometric mechanism of Pareto compression, §6.5 verifies the condition dependence of this mechanism through ablation, and §6.6 separates initialization and search-budget confounds through robustness controls.

6.2. Squint-Aware Special Regime

The squint-minimized greedy solver (G-SM) improves NESZ radiometric quality (f_3) relative to the profit-only baseline (G-BL) by selecting observation times that minimize squint angle within each visibility window. Across all 200 scenarios, this produces a large positive effect on f_3 (Cliff's $\delta = +0.99$, $p \approx 1.5 \times 10^{-34}$, on a per-scenario paired basis: 199 of 200 scenarios improve) at a substantial aggregate cost in coverage (Cliff's $\delta = -1.0$, $p < 10^{-34}$, on f_1^*; raw f_1 is used here because G-BL yields constant $f_1^* = 1.0$ by definition, for which the normalized-profit comparison degenerates to a one-sample test against a fixed value). The trade-off, however, is strongly density-dependent. In sparse scenarios (S1, $N = 20$), G-SM drops to $f_1^* = 0.51 \pm 0.16$ (-49% vs. G-BL at 1.00) while improving f_3 from 0.497 ± 0.044 to 0.722 ± 0.058 (+45%). In dense scenarios (S4, $N = 500$), the f_1 cost escalates: G-SM achieves $f_1^* = 0.37 \pm 0.12$ (-63%), with f_3 rising from 0.675 ± 0.074 to 0.786 ± 0.046 (+16%). Fig. 2 visualizes this density-dependent trade-off: the Locally Estimated Scatterplot Smoothing (LOESS) trend in panel (c) identifies a transition region around $N = 100$ where the f_1 penalty grows rapidly. The radiometric advantage narrows with density (+45% at S1 to +16% at S4) as tightly packed time windows constrain how far each acquisition can be moved toward the squint-minimizing time, while the coverage cost remains large at every density (G-SM schedules, on average, 44% of the tasks G-BL selects at S1, falling to $\sim$20–27% at $N \geq 100$; these are means of the per-scenario scheduled-task-count ratios, the profit-side analogue of which appears in Table 3).

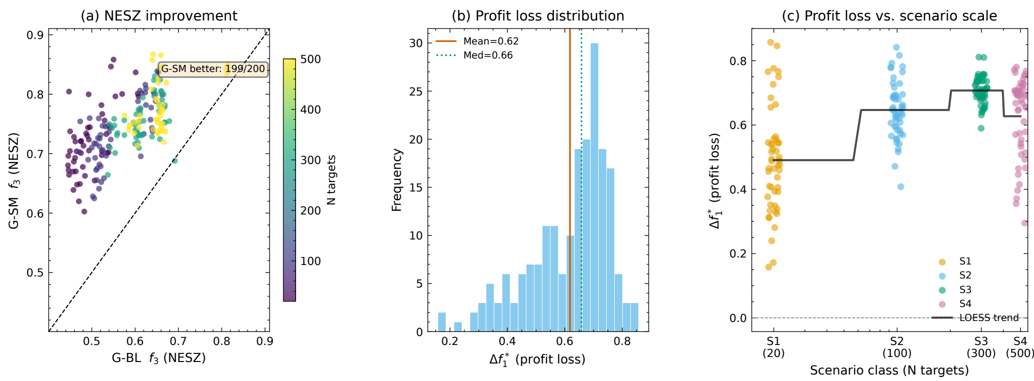

Figure 2: Effect of squint-aware scheduling: (a) f_3 scatter; (b) Δf_1^* histogram; (c) Δf_1^* vs. N with LOESS trend

6.3. Scale Sensitivity

The gap between MOEA-2 and G-BL varies systematically with scenario density (Fig. 3, Table 5). MOEA-2's coverage deficit relative to G-BL is confined to sparse densities: from $f_1^* = 0.86 \pm 0.14$ at S1 ($N = 20$) it rises to 0.985 ± 0.03 at S2 ($N = 100$), and reaches 0.996–0.999 at $N \geq 150$ (S7–S8, S3), remaining within 1% of G-BL at S4 (0.996). The f_2 improvement over G-BL follows the same contraction, falling from +33.7% (S1) to +3.5% (S2) and to +1.3% or less at $N \geq 150$. The proportion of scenarios where MOEA-2's coverage falls measurably short of G-BL ($f_1^* < 0.95$, i.e., a coverage deficit exceeding 5%) collapses from 80% (S1) to 10% (S2) and to 0% at $N \geq 150$. The coverage cost of multi-objective optimization is therefore a sparse-density phenomenon: between $N = 20$ and $N = 100$ the solver transitions from actively trading off coverage for quality to operating at the greedy baseline, and beyond $N = 150$ the deficit is effectively zero.

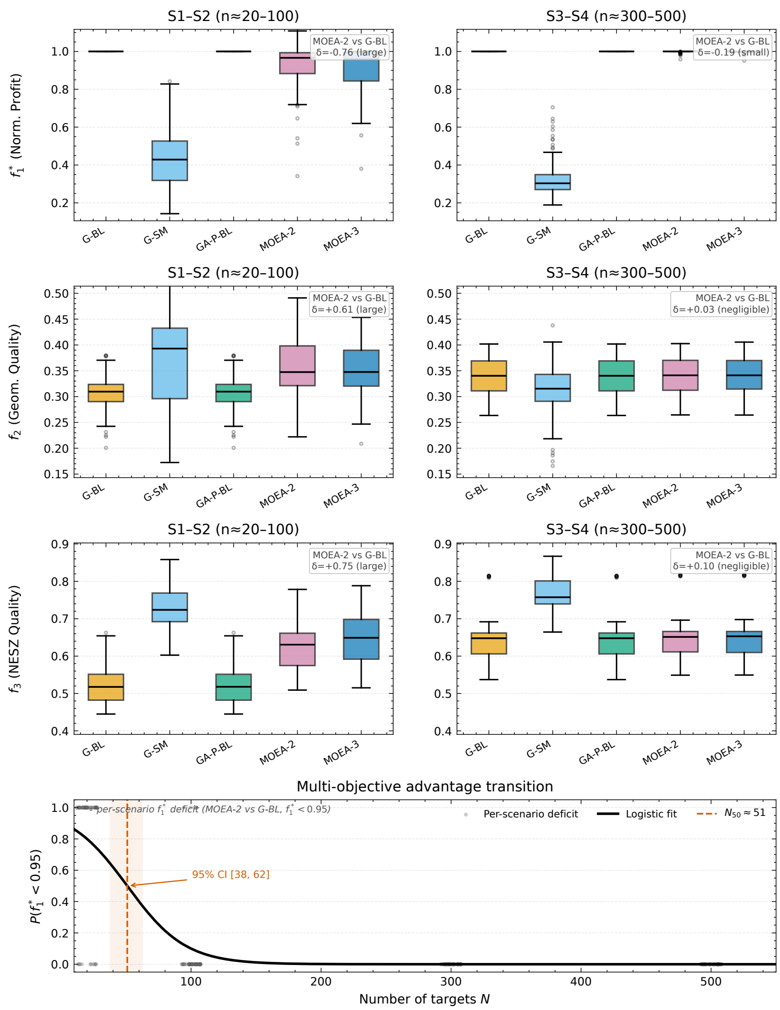

Figure 3: Scale sensitivity: (top) per-objective distributions of normalized profit f_1^*, geometric quality f_2, and radiometric quality f_3 for sparse (S1–S2) and dense (S3–S4) scenario classes; (bottom) logistic fit to the proportion of scenarios with a coverage deficit ($f_1^* < 0.95$), with jittered per-scenario indicators.

Table 5: Scale sensitivity of MOEA-2 vs. G-BL. f_2 improvement is the mean of per-scenario ratios (+34% at S1 on this basis; the ratio of pooled means gives +33%). S7 and S8 are intermediate scale points run for MOEA-2 and the two greedy baselines only (§5.4); classes are ordered by target count N.

Class	N	f_1^* (MOEA-2)	f_2 improvement	% $f_1^* < 0.95$
S1	20	0.86 ± 0.13	+33.7% (±14.3)	80%
S2	100	0.98 ± 0.03	+3.6% (±3.6)	10%
S7	150	1.00 ± 0.01	+1.3%	0%
S8	200	1.00 ± 0.00	+0.8%	0%
S3	300	1.00 ± 0.00	+0.4% (±0.2)	0%
S4	500	1.00 ± 0.01	+0.3% (±0.2)	0%

Fitting a logistic regression to the proportion of scenarios with $f_1^* < 0.95$ across the six density points ($N = 20, 100, 150, 200, 300, 500$) yields a transition midpoint $N_{50} \approx 50$ targets. This estimate is weakly constrained: only the two sparse densities ($N = 20, 100$) carry informative deficit proportions (80% and 10%), while the four denser points are all at most 1%, so the bootstrap confidence interval is wide ($N_{50} \in [30, 62]$). N_{50} should therefore be read as a coarse indicator that the transition is confined to $N \in [20, 100]$, not as a precise threshold. Beyond $N \approx 150$, no scenario shows a coverage deficit exceeding 5%. The transition between a measurable and a negligible multi-objective advantage therefore lies in the $N \in [20, 100]$ band, and beyond $N = 150$ the advantage is effectively absent. The f_3 row of Fig. 3 shows the same pattern on the radiometric objective: at sparse density MOEA-3 separates visibly from G-BL and G-SM sits in its own high-f_3 regime, whereas the dense panels show all solvers converging to nearly identical distributions, consistent with Table 3.

6.4. Mechanism of Pareto Compression

Phenomenon 1: Front compression. Fig. 4 visualizes the Pareto front on a representative S4 scenario. The front contains 4.7 ± 4.4 MOEA-3 non-dominated solutions (MOEA-2: 1.6 ± 0.9) but is compressed: the found front spans only about 0.08 in f_1^* and 0.014 in f_2, against a total span of $\Delta f_2 \approx 0.14$ in sparse S1 scenarios, confirming the scale-dependent saturation documented in §6.3. This compression is a property of the found front under G-BL-seeded search rather than of the full feasible trade-off surface: the squint-minimizing heuristic G-SM reaches feasible high-f_3 operating points

($f_3 \approx 0.79$ at S4, §6.2) outside the MOEA-3 surface, so the dense-regime indistinguishability is established near the full-coverage working point rather than over the entire feasible surface. In dense regimes, solver convergence to lower-squint geometries combined with schedule saturation limits quality differentiation regardless of solver strategy. The MOEA-2 and MOEA-3 fronts are nearly coincident at this density, indicating that adding f_3 as a third objective produces little additional differentiation in the dense setting.

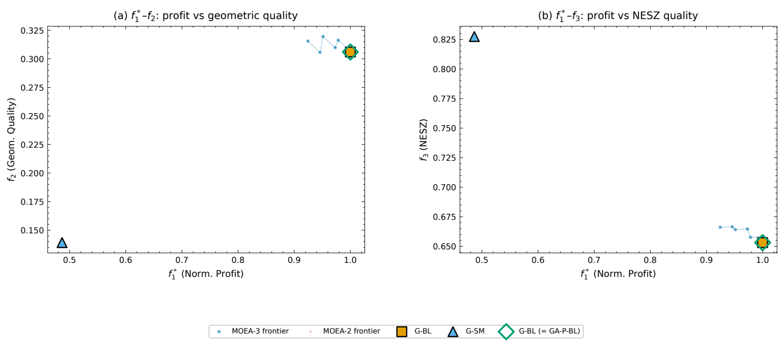

Figure 4: Pareto front on a representative S4 scenario ($N = 500$): MOEA-3 frontier with MOEA-2 overlay and baseline solver markers. Panel (a): f_1^*–f_2; panel (b): f_1^*–f_3.

The f_1^*–f_2 projection (panel a) shows an extremely compressed surface: even for the S4 scenario with the widest f_1^* spread, the frontier spans only $f_1^* \in [0.92, 1.00]$ and $f_2 \in [0.306, 0.320]$ — a near-vertical segment at full coverage. Overlaying the single-point baseline solutions shows that G-BL sits at the "full-coverage, moderate-quality" corner of the frontier, while G-SM occupies a qualitatively different region ($f_1^* \approx 0.37$, $f_3 \approx 0.79$) that lies outside the MOEA-3 Pareto surface, a consequence of G-SM's aggressive squint minimization.

Phenomenon 2: Density-dependent third-objective value. The differentiation of MOEA-3 from MOEA-2 is density-dependent but modest: at S1 the physics-aware solver attains $f_3 = 0.698 \pm 0.039$ versus 0.657 ± 0.040 for MOEA-2 (+6.2% under the reported knee rules, paired Wilcoxon $p = 5.5 \times 10^{-13}$; the matched-pairs sign statistic is +0.88 on a per-scenario basis (47 of 50 scenarios), corresponding to a conventional Cliff's δ of +0.57), at a small f_2 cost (0.376 ± 0.050 vs. 0.393 ± 0.055, -4%, $p = 0.017$). This margin is conditional on the knee-point rule: because MOEA-3's knee maximizes $f_1 + f_2 + f_3$ while MOEA-2's knee maximizes $f_1 + f_2$ (§ 5.3), re-selecting both fronts under a matched radiometric-aware rule gives a controlled third-objective margin of +4.9% ($p = 2.3 \times 10^{-11}$, 47 of 50), whereas a radiometric-blind rule gives +0.05% ($p = 0.86$); the headline comparison's remaining gap thus conflates the objective function with knee selection, and the objective itself contributes the +4.9% controlled margin that only materializes under a radiometric-aware operating point. This +6.2%/+4.9% is robust to the reference-direction count: rerunning MOEA-3 at S1 with Das–Dennis $p = 4$ ($H = 15$, comparable in magnitude to MOEA-2's $H = 13$ instead of the default $H = 91$) yields $f_3 = 0.702 \pm 0.041$ (50 scenarios, same pop=100, gen=200, G-BL hot-start), statistically indistinguishable from the $p = 12$ value 0.698, so the third-objective gain is not an artifact of the 7× larger reference-direction set; at S4 the means coincide (0.680 ± 0.073 for both), though MOEA-3 retains a modest per-scenario f_3 advantage (33 of 50, paired sign statistic +0.32, $p = 4.5 \times 10^{-3}$; conventional Cliff's $\delta = +0.03$, i.e. negligible in distribution although consistently signed). The third objective thus contributes a modest, consistent f_3 gain at sparse density rather than a qualitatively distinct regime: under the elevation-plane decomposition of §3.2, MOEA-2 already attains high f_3 post-hoc, so the marginal value of explicitly optimizing f_3 is bounded, and the principal quality contrast with the profit-only baseline remains MOEA-3's +40% f_3 gain at S1. At operational densities the advantage disappears because coverage pressure locks the selection, not because the underlying f_2–f_3 trade-off vanishes—the within-schedule correlation remains negative but is strongest at sparse density (about −0.6 to −0.7), settling to a moderate ≈ -0.2 to -0.15 at higher density (§6.4).

The HV ranking (Table 4) differs from the coverage ranking (Table 3): among the three tabulated solvers, G-BL ranks first in tabulated f_1^* but third in HV, while the NSGA-III variants rank first and second in HV (MOEA-3, then MOEA-2), despite their lower coverage ranking. Notably, G-SM — omitted from the table because its single-point HV is dominated by point location rather than front quality — actually attains the highest HV of any solver (0.2341) from its low-coverage, high-f_3 operating point ($f_1^* \approx 0.37$, $f_3 \approx 0.79$ at S4), so the HV ranking would invert if it were included. This is precisely the point-count confound: HV rewards an extreme single point more than a dense front near full coverage. One sparse-density MOEA solution exceeds $f_1^* = 1.00$ (one of 200 scenarios, +10.8% above G-BL), so G-BL is a near-optimal rather than a strict coverage anchor. This reversal is driven by point-count asymmetry: single-objective solvers contribute one point each to the HV reference set, while MOEA-3 contributes 33.8 ± 8.4 non-dominated points at S1 (MOEA-2: 6.8±2.1), a cardinality gap that persists, less severely, at S3–S4. HV therefore conflates front quality with front cardinality, and the ranking should not be interpreted as a standalone quality judgment. The core phenomenon—front compression—is independent of HV ranking.

Mechanism: geometric coupling and lower-squint convergence. Because θ and ψ_{sq} are not independent but both derived from the same LOS vector $\mathbf{l} = (l_x, l_y, l_z)$ (§ 3.1), the per-task f_2 and f_3 contributions share geometric structure. To separate definitional coupling from scheduler-induced correlation, we compute the analytical null distribution by treating θ and ψ_{sq} as independently and uniformly distributed within the accessible envelope ($\theta_{\mathrm{elev}} \in [15^\circ, 50^\circ]$, $|\psi_{\mathrm{sq}}| \leq 45^\circ$, slightly wider than the C1 constraint $[18^\circ, 47^\circ]$ to provide a conservative null baseline; the same experimental squint envelope as §3.1, not a specific platform specification) and evaluating the Pearson correlation of $f_{2,i} = \sin\theta_{\mathrm{elev},i}\cos\psi_{\mathrm{sq},i}$ and $f_{3,i} = \cos^3\xi_i$ under independent angles. Two independent baselines give the same order of magnitude: a permutation null that independently shuffles the observed θ_{elev} and ψ_{sq} marginals within each density class gives $r_{\mathrm{null}} = -0.60$ (95% spread ± 0.02 across 20 shuffles per class), and Monte Carlo sampling over the uniform box $\theta_{\mathrm{elev}} \in [16^\circ, 41^\circ]$, $|\psi_{\mathrm{sq}}| \leq 45^\circ$ gives $r_{\mathrm{MC}} = -0.28$ ($N_{\mathrm{MC}} = 2 \times 10^6$). We adopt the empirical-marginal permutation value as the factorized null:

$$r_{\mathrm{null}} = \frac{\mathrm{Cov}(f_2, f_3)}{\sqrt{\mathrm{Var}(f_2)\,\mathrm{Var}(f_3)}} \approx -0.60, \tag{6}$$

where the independence of θ_{elev} and ψ_{sq} under the null factorizes the moments, e.g., $E[f_2 f_3] = E[\sin\theta_{\mathrm{elev}}\cos^3\theta_{\mathrm{elev}}]\,E[\cos^4\psi_{\mathrm{sq}}]$. The *negative* sign exposes a structural conflict, not redundancy: in the θ_{elev}-only limit (fixed squint), $f_{2,i} \propto \sin\theta_{\mathrm{elev}}$ and $f_{3,i} \propto \cos^3\theta_{\mathrm{elev}}$ are strictly anti-monotonic ($r_\theta = -0.997$ in Monte Carlo), since f_2 favors high incidence while f_3 favors low incidence; in the ψ_{sq}-only limit (fixed θ_{elev}), both depend on $\cos\psi_{\mathrm{sq}}$ positively ($r_\psi = +0.997$), since low squint improves both. The wide-envelope Monte Carlo value (-0.28) is much weaker than the broadside limit because the positive ψ-common-mode dilutes the θ_{elev} conflict.

Empirical correlation and lower-squint convergence. Pooled across all scheduled tasks from the 200 scenarios, the per-task empirical correlation between f_2 and f_3 within the solver-selected schedules is negative at every density, and it is *strongest at sparse density* rather than strengthening as density rises: for the full-physics MOEA-3 (variant A) the pooled correlation is -0.63 at S1 and falls to -0.15 (S2), -0.13 (S3), and -0.23 (S4). Because the pooled statistic mixes between-schedule and within-schedule variance, we also computed per-schedule correlations (Pearson r within each solver-selected schedule, one per scenario): the per-schedule mean is -0.72 at S1 and $-0.15/-0.16/-0.18$ at S2/S3/S4 (Fisher-z means $-1.09/-0.16/-0.16/-0.19$, i.e. the mean of $\operatorname{atanh}(r)$ over the 50 schedules per class), and individual schedules span a wide range (S1 $r \in [-0.97, -0.03]$, entirely negative), so the pooled value is representative but not uniform across schedules. The θ_{elev}-conflict therefore persists inside sparse schedules rather than being reversed. All solvers select geometries shifted toward lower squint than the visibility envelope (mean $|\psi_{\mathrm{sq}}|$ 16–24°; §3.1), and the schedule correlation is strongly density-*dependent*: at sparse density the full-physics schedules match the factorized, independent-angle null (-0.63 versus $r_{\mathrm{null}} \approx -0.60$), whereas at operational density they weaken to about -0.2 to -0.15. The third objective's class-mean indistinguishability at high density is therefore an empirical convergence effect—not a mathematical identity between f_2 and f_3: a trace per-scenario f_3 advantage of MOEA-3 over MOEA-2 persists even at S4 (paired Wilcoxon $p = 4.5 \times 10^{-3}$, 33 of 50 scenarios), coexisting with negligible class-mean f_3 margins.

Envelope versus schedule correlation. Across the visibility-window grid (all window time-steps from 20 scenarios, five per density class S1–S4 (seeds 0–4), at 10 s resolution; 7.6×10^4 geometry samples), evaluating the same objective decomposition gives $\operatorname{corr}(f_2, f_3) \approx -0.36$ at every density (-0.35 at S1 to -0.37 at S2–S3). This envelope value is already weaker than the factorized null $r_{\mathrm{null}} \approx -0.60$ (Eq. 6): real visibility geometry couples θ_{elev} and ψ_{sq} through the LOS vector (both grow toward pass edges), adding a positive common-mode term that dilutes the θ_{elev} conflict. The reported envelope therefore spans squints only up to the C7 squint limit used at window generation ($|\psi_{\mathrm{sq}}| \leq 45°$); the scheduled geometries concentrate at much smaller squints.

The solver-selected schedules move the correlation in opposite directions at the two density regimes: at sparse density they restore the full independent-angle conflict (-0.63, matching the -0.60 null)—the knee solutions traverse the θ_{elev}–ξ trade-off across the selected tasks instead of tracking the envelope's co-varying geometry; at high density, coverage packing weakens the correlation below the envelope value (-0.15 to -0.23). The sparse strengthening is an active effect of the physical objectives, not of envelope geometry: the visibility envelope alone supplies only the weaker -0.36 correlation.

Ablation studies identify which objective component drives this pattern. Under variant B (squint removed from the objectives: $f_2 = \sin\theta_{\mathrm{elev}}$, $f_3 = \cos^3\theta_{\mathrm{elev}}$ only), the sparse schedule correlation collapses to $r = -0.23$, close to the envelope value -0.36; under variant C (incidence removed: squint-only objectives $f_2 = \cos\psi_{\mathrm{sq}}$, $f_3 = \cos^3\psi_{\mathrm{sq}}$) it remains strong ($-0.65$), indistinguishable from the full-physics variant A (-0.63); under variant D (all physical objectives removed) it falls to -0.13 at S1 and -0.07 at S2. The schedule-level conflict is thus realized by any configuration that steers squint: variants that see ψ_{sq} (A and C) select configurations spanning the full θ_{elev} trade-off, while variants blind to ψ_{sq} (B and D) produce schedules whose within-schedule geometry sits at the weak envelope level. In the sparse regime the squint-aware objectives thus retain solutions that traverse the f_2–f_3 trade-off; without that steering the solver maximizes coverage alone and temporal packing leaves the geometry near envelope statistics. At higher density the full-physics residual correlation also settles at a moderate level (about -0.2 to -0.15), so the ablation contrast narrows; the third objective's value is lost not because the conflict vanishes but because coverage pressure leaves no room to traverse the trade-off and the f_3 margins between solvers become negligible. A mechanical alternative—that the dense correlations weaken because the realized f_2 value range compresses—is rejected by the data: the pooled within-schedule f_2 standard deviation is effectively constant across density classes (0.16 at S1–S3, 0.14 at S4 for variant A; pooled range 0.65 in every class). Consistent with this scale-dependent saturation, the per-density hypervolume contracts at higher density: at S4 the pooled HV is 0.25 (MOEA-2) and 0.17 (MOEA-3) of its S1 value, i.e. the S4/S1 HV ratio is well below unity, reflecting the near-point front compression documented in §6.4 (S4 fronts span only $\Delta f_1^* \approx 0.06$) rather than a degradation of solution quality. We caution that the Pareto fronts reported here are best-known approximations obtained via NSGA-III; whether they represent the true Pareto-optimal set is unknown without exact methods, which remain intractable for $N \geq 20$ in this problem class.

6.5. Physical-Objective Ablation

To quantify the contribution of incidence-angle (θ) and squint-angle (ψ_{sq}) modeling to solver performance, we repeat the MOEA-3 experiment with four objective-function variants:

A. Full physical (baseline) $f_2 = \frac{1}{|\mathcal{S}|}\sum \sin\theta_{\mathrm{elev},i}\cdot\cos\psi_{\mathrm{sq},i}$, $f_3 = \frac{1}{|\mathcal{S}|}\sum \cos^3\xi_i$

(identical to the standard configuration in § 6.4).

B. No squint $f_2 = \frac{1}{|\mathcal{S}|}\sum \sin\theta_{\text{elev},i}$, $f_3 = \frac{1}{|\mathcal{S}|}\sum \cos^3\theta_{\text{elev},i}$ (removes ψ_{sq} component; θ_{elev} is the elevation-plane incidence angle, $\cos\theta_{\text{elev}} = \cos\xi/\cos\psi_{\text{sq}}$).

C. No incidence $f_2 = \frac{1}{|\mathcal{S}|}\sum \cos\psi_{\text{sq},i}$, $f_3 = \frac{1}{|\mathcal{S}|}\sum \cos^3\psi_{\text{sq},i}$ (removes θ_{elev} component).

D. No physics $f_2 = f_3 = 1$ (constant per-task objective; solver receives no geometric information; f_2, f_3 shown in Table 6 are post hoc diagnostics computed from the actual observation geometry of the resulting schedule). Variant D serves as a null-model control: it tests whether the NSGA-III solver's performance is attributable to the physical content of f_2 and f_3, or merely to the presence of additional objectives that structure the search.

All four variants use identical NSGA-III parameters (population = 100, generations = 200, Das-Dennis $p = 12$), the same scenario files, and the same G-BL hot-start injection. Only the objective functions differ, making this a controlled ablation rather than a retuning experiment. Within-solver ablation with identical NSGA-III parameters and the same G-BL seed injection controls confounding factors sufficiently that the Wilcoxon signed-rank test is appropriate for paired comparison of objective-function variants. Each variant completes six scenario classes $\times$ 50 seeds = 300 runs (total: 1,200 runs across A–D); Table 6 reports the density-regime classes S1–S4. The two remaining classes stress non-density axes: S5 holds density at $N = 20$ but spreads target positions along-track over $5°/15°/25°/35°/45°$ (a squint-spread sensitivity group), and S6 packs the 20 targets into 2–10 clusters with narrow windows (60–180 s; a C3 transition-stress group). Degradation is reported as $\Delta\% = (\mathrm{A}-\mathrm{V})/\mathrm{A}\times 100$, where A is the full physical baseline and V is the variant; negative values indicate the variant outperforms the baseline on f_1^*.

Table 6: Per-class ablation results. Bold: $p < 0.005$ (Wilcoxon signed-rank vs. A; consistent with the Friedman–Wilcoxon framework of §5.3).

Class	**V**	f_1^*	f_2	f_3	f_1/n	$n_{\mathbf{sel}}$	$\Delta f_1^*\%$
S1 ($N = 20$)	A	0.850 ± 0.123	0.376	0.698	5.99 ± 0.73	15.6 ± 2.6	(baseline)
	B	0.852 ± 0.152	0.353	0.574	6.01 ± 0.83	15.7 ± 3.1	-0.8%
	C	0.845 ± 0.095	0.375	0.703	5.86 ± 0.80	16.0 ± 2.2	$+0.1\%$
	D	1.002 ± 0.011	0.302	0.525	5.65 ± 0.66	19.5 ± 0.7	$\mathbf{-18.0}\%$
S2 ($N = 100$)	A	0.976 ± 0.040	0.329	0.597	5.58 ± 0.42	62.7 ± 19.3	(baseline)
	B	0.990 ± 0.022	0.321	0.570	5.55 ± 0.37	63.6 ± 19.1	$\mathbf{-1.2}\%$
	C	0.977 ± 0.033	0.328	0.599	5.57 ± 0.40	62.8 ± 19.3	$+0.0\%$
	D	1.000 ± 0.000	0.317	0.562	5.54 ± 0.37	64.2 ± 18.8	$\mathbf{-2.0}\%$
S3 ($N = 300$)	A	0.998 ± 0.005	0.344	0.623	5.63 ± 0.30	154.2 ± 39.9	(baseline)
	B	0.998 ± 0.005	0.343	0.618	5.63 ± 0.29	154.2 ± 40.0	$+0.0\%$
	C	0.998 ± 0.004	0.344	0.623	5.63 ± 0.30	154.2 ± 39.9	$+0.0\%$
	D	1.000 ± 0.000	0.342	0.616	5.63 ± 0.29	154.5 ± 39.9	$\mathbf{-0.2}\%$
S4 ($N = 500$)	A	0.997 ± 0.008	0.338	0.680	5.83 ± 0.44	142.3 ± 72.1	(baseline)
	B	0.999 ± 0.003	0.337	0.677	5.84 ± 0.44	142.5 ± 72.0	-0.1%
	C	0.998 ± 0.006	0.337	0.680	5.83 ± 0.44	142.4 ± 72.0	$+0.0\%$
	D	1.000 ± 0.000	0.336	0.675	5.83 ± 0.45	142.7 ± 71.9	$\mathbf{-0.2}\%$

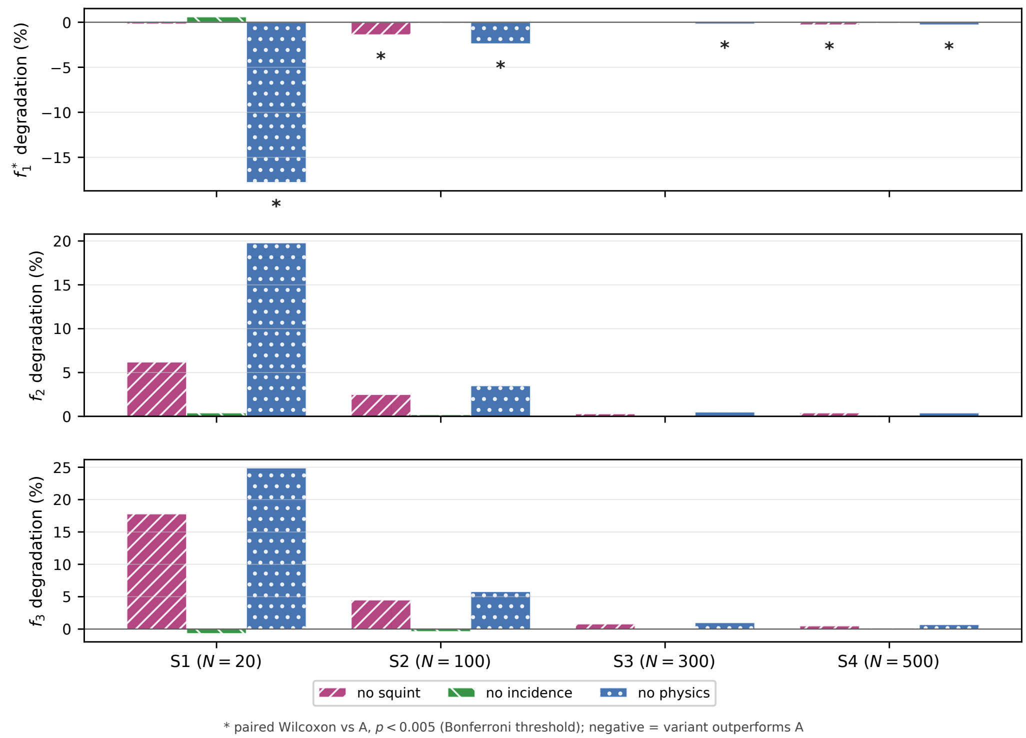

Figure 5: Ablation results across objective-function variants B–D and scenario classes S1–S4. Bars show degradation relative to the full-physics baseline A for normalized profit f_1^* and post hoc geometric-quality metrics f_2 and f_3; negative values indicate that a variant outperforms A. Asterisks mark paired Wilcoxon comparisons against A with $p < 0.005$ (Bonferroni threshold).

The ablation results (Table 6, Fig. 5) reveal that the influence of physical objective modeling is concentrated in sparse scenarios and vanishes as scenario density increases — a pattern consistent with our scale-sensitivity analysis (§ 6.3).

At S1 ($N = 20$), only variant D (no physics) differs significantly from baseline A at the manuscript's $p < 0.005$ threshold: its f_1^* rises to 1.002 (-18.0%, $p = 3.4 \times 10^{-9}$) by scheduling 3.9 more tasks (19.5 vs. 15.6) at a substantial radiometric cost, with post hoc f_3 falling from 0.698 to 0.525 (-25%). Variant B (no squint) leaves f_1^* unchanged (-0.8%, $p = 0.46$) but lowers f_3 to 0.574 (-18%) and f_2 to 0.353 (-6.2%, $p = 1.1 \times 10^{-5}$): without squint penalization the solver tolerates higher-squint views, sacrificing the radiometric quality that the full-physics baseline purchases in the sparse regime. Variant C (no incidence) produces no measurable change in any reported metric. This inertness is consistent with the window geometry: the elevation-plane incidence angle varies by only a few degrees across a visibility window (mean $\sim 1.5^\circ$ in the S1 scenarios), whereas the squint angle sweeps ~ 20–35°, so the θ_{elev} factor in f_2/f_3 acts as an approximately common scale within each window; removing it preserves the within-window task ordering and hence the selected schedule. The incidence conflict of §6.4 therefore operates through the across-window/aggregate geometry rather than through per-window selection. Thus the physical content of the objectives acts on sparse-regime outcomes chiefly through radiometric quality (f_3) and through the full-removal coverage contrast, while partially removing either geometric component leaves coverage unaffected.

At S2 ($N = 100$), variants B and D remain significantly different from A but the effects are small (-1.2%, $p = 5.3 \times 10^{-4}$ and -2.0%, $p = 1.7 \times 10^{-6}$): removing squint or all physics still changes coverage modestly. Variant C remains indistinguishable from A.

At S3–S4 ($N \geq 300$), every variant lies within 0.2% of the baseline for f_1^* and within 0.6% for f_2; D's paired-test significance reflects a consistent but negligible effect (-0.2%). This numerical invariance is the direct signature of the scale-dependent saturation discussed in § 6.3: once the density of candidate observation windows saturates the schedule, the objective function only penalizes infeasible solutions rather than differentiating among feasible ones. In this regime, the NSGA-III solver acts as a constraint-filtered profit maximizer regardless of whether f_2 and f_3 carry geometric meaning (A) or serve as constant placeholders (D).

The two robustness classes replicate this pattern away from the density axis. On S5 (along-track target spread) and S6 (narrow-window clusters), the full-removal variant D is significantly worse than A in f_2 (-11.2%, $p = 1.6 \times 10^{-13}$ and -11.1%, $p = 1.8 \times 10^{-15}$) and in f_3 (-6.1% and -6.3%), C is indistinguishable from A (f_2 $+0.4\%$, $p = 0.33$ and -0.6%, $p = 0.48$), and B shows small significant differences ($+3.2\%$, $p = 0.046$ and $+4.6\%$, $p = 1.2 \times 10^{-5}$ in f_2). The coverage price of physics persists here as well: D attains 17% (S5) and 8% (S6) more raw profit than A, i.e. objective-driven selection sacrifices coverage in sparse configurations regardless of whether the stressor is density, target spread, or window clustering.

Taken together, the physical quality objectives add value mainly in the sparse regime, and primarily through radiometric quality: at S1 the full-physics A attains 33% higher f_3 than the no-physics D (0.698 vs. 0.525;

D lies 25% below A on the common-degradation basis) at an 18% coverage disadvantage, and removing only the squint component (B) already sacrifices 18% of f_3 while preserving f_1^*. The coverage effect of partially removing the objectives is minimal in every regime, and by $N \geq 300$ the choice of physical objectives is numerically irrelevant to outcomes.

6.6. Robustness and Calibration

The mechanism above must be separated from initialization and search-budget effects. We therefore use four control experiments.

Hot-start dependence.. On 10 S1, 4 S2, 5 S3, and 5 S4 scenarios (3 seeds each), random-initialized MOEA-2 yields f_1^* deficits of approximately -0.393 ± 0.251, -0.551 ± 0.060, -0.684 ± 0.050, and -0.475 ± 0.067 (mean $\pm$ SD across runs) relative to hot-start MOEA-2, respectively. G-BL seeding therefore materially improves coverage convergence at every tested density. The deficit peaks at the dense S3 class (-0.68 at $N = 300$), is smallest at the sparsest S1 (-0.39 at $N = 20$), and moderates at the densest S4 (-0.48 at $N = 500$), so high target density does not by itself make random initialization sufficient. This constrains the coverage-convergence claim: proximity to G-BL is initialization-dependent. The initialization effect is not confined to coverage: random-init f_2 is actually higher — i.e. better geometric quality, under the convention of Table 3 — at sparse density (0.505 vs. 0.395 at S1, +28%; 0.460 vs. 0.320 at S2, +44%), marginally higher at S3 (0.349 vs. 0.339, +3%), and only lower at the densest S4 (0.347 vs. 0.380, −9%). Random initialization therefore attains comparable or better per-task geometric quality at S1–S3 at a large coverage cost, and the G-BL anchor's material benefit lies in coverage, not geometry; its geometric advantage is confined to the densest regime. The coverage convergence is thus a G-BL initialization effect rather than an artifact of the MOEA framework. Two caveats qualify this control: the deficit sample is small (10/4/5/5 scenarios per class $\times$ 3 seeds, and 30 runs for the variant-D control, 15 per class), the scenarios were selected as the first seeds of each density class in manifest order (subconfig A, seeds 00 onward), the hot and random runs share the same scenario seeds, so each deficit is a within-seed paired difference rather than an independent-sample contrast, and the comparison is expressed on the G-BL-normalized f_1^* scale, on which the greedy anchor itself reads exactly 1.0 by construction; both argue for reading the deficit magnitudes as indicative of the initialization effect rather than as precise coverage bounds.

Perturbation sensitivity.. A sweep over $\sigma \in \{0.1, 0.3, 0.5, 0.7\}$ on MOEA-3 (10 S3 scenarios per value, fixed solver seed) leaves mean f_2 and f_3 essentially unchanged (mean $f_2 \approx 0.343$ and mean $f_3 \approx 0.610$ at every level, with level-to-level variation of at most $\sim 1 \times 10^{-3}$; MOEA-2 shows the same flatness at $f_2 \approx 0.343$ and $f_3 \approx 0.609$), and coverage stays flat at 200/300 selected tasks (199.5 at $\sigma = 0.7$). The per-task f_2–f_3 coupling reported in § 6.4 is therefore a property of the schedule geometry rather than of the perturbation magnitude, and the indistinguishability persists across $\sigma = 0.1$–0.7 rather than being a byproduct of the $\sigma = 0.5$ perturbation level. This sweep was run on S3 scenarios only (the dense transition class); the perturbation flatness is therefore directly established at high density and is not independently assessed in sparse regimes. All control runs use the same Das–Dennis partition ($p = 12$, giving $H = 91$ reference directions for three objectives and $H = 13$ for two) as the headline experiments.

Random-search calibration.. On five S1 scenarios, 5,000 random task selections per scenario produce mean $f_1^* = 0.527 \pm 0.003$ and a 90th percentile of 0.930 ± 0.008; the best sample reaches 1.000 — the G-BL profit, so random sampling never exceeded the greedy baseline on these scenarios. Because the random sampler is unconstrained and does not enforce C2–C4, this is an empirical bound on the tested batch rather than a construction guarantee. High coverage under random sampling therefore occurs only by chance; the MOEA's gains reflect structured search.

Random-init control.. A random-init variant-D control (15 S3 and 15 S4 runs, no G-BL hot start) also converges to low squint, with $f_2 = 0.349 \pm 0.016$ at S3 and 0.347 ± 0.015 at S4, statistically indistinguishable from the hot-started D solver (0.342 ± 0.026 at S3, 0.336 ± 0.039 at S4) on this per-task mean at both densities. This f_2 agreement masks a coverage collapse: the random-init D attains only $f_1^* = 0.319 \pm 0.044$ and selects 70.3 tasks at S3 (0.522 ± 0.068 and 117.8 at S4), against near-full coverage for the hot-started D, so the inference applies to the per-task geometric mean on much smaller schedules rather than to coverage. The lower-squint convergence is therefore objective-intrinsic rather than a hot-start residue: the G-BL seed inserts tasks at their earliest feasible time and thus biases hot-started runs toward the higher squint of the greedy schedule, most visibly at high density. The A–D invariance itself already holds at standard budget (Table 6: $\Delta f_1^* \leq 0.2\%$ at S3–S4), a physically forced consequence of coverage pressure locking the selection at high density (§ 6.4); the convergence is therefore not a budget artifact.

Search-budget control.. To verify directly that the A–D invariance does not depend on the search budget, we reran the full-physics (A) and no-physics (D) variants at double budget (population 200, 400 generations) on five S3 and five S4 scenarios under the current solver code. The A–D difference in f_1^* remains negligible (-0.12% at S3, -0.44% at S4; both variants reach full coverage), corroborating the standard-budget ablation and confirming that the invariance reflects coverage pressure at high density rather than the budget expended. Quality is likewise budget-stable: comparing the double-budget full-physics runs against the standard-budget runs on the same ten scenarios shifts f_1^* by only -0.05% (S3) and -0.35% (S4), f_2 by $+0.64\%/+0.52\%$, and f_3 by $+1.64\%/+0.97\%$ (all below 2%), so the reported sub-2% dense-regime quality margins are not an artifact of premature convergence at the tested budget; a convergence trace for the 200-generation frontier is not reported, and the quality claims are understood as conditional on the tested budget. These controls collectively favor the lower-squint convergence explanation over hot-start residue, noise-level sensitivity, or random chance, while not ruling out all algorithm-specific effects.

7. Discussion

7.1. Summary of Findings

At $N = 20$ the three dimensions form a genuine trade-off surface — G-BL retains $f_1^* = 1.00$ at a moderate f_3 (0.497), MOEA-2 reaches the highest f_2 (0.393) alongside a f_3 of 0.657, and MOEA-3 trades f_2 (-4% vs. MOEA-2) for the highest f_3 (0.698, +6.2% vs. MOEA-2 under the reported knee rules) — so no solver dominates on more than one dimension, whereas at $N \geq 300$ all solvers converge to the G-BL anchor and the greedy baseline is operationally sufficient. Our experiment across 200 systematically varied scenarios yields five interconnected findings: **(1)** the quality advantage of multi-objective optimization saturates with target density (+33.7% in f_2 at $N = 20$ falling to +0.4% at $N = 500$); **(2)** the third objective is density-dependent rather than redundant: per-task f_2–f_3 correlation within schedules stays negative at every density, strongest at sparse density (about -0.63 pooled, -0.72 per-schedule, at S1 for the full-physics solver) and settling to a moderate ≈ -0.2 to -0.15 at higher density, and MOEA-3 separates from MOEA-2 in f_3 at sparse density (+7% at $N = 20$) with the advantage shrinking to $\lesssim 0.2\%$ at $N \geq 300$, confirming the coverage-pressure mechanism detailed in § 6.4. Taken together, these results establish that the Pareto front, while offering multiple quality-coverage configurations, is compressed by the geometric coupling above: the narrow f_2 range limits practical differentiation between solutions. **(3)** G-SM occupies a distinct high-radiometric-quality regime at substantial density-dependent coverage cost; **(4)** solver outcomes become invariant to the choice of physical objectives at $N \geq 300$ (ablation study); and **(5)** coverage convergence toward the G-BL anchor is conditional on the hot start, not a structural property. The first four reflect geometric properties of the single-pass observation geometry (finding (2) being an empirical convergence effect under those properties, § 6.4); the fifth reflects solver initialization.

The Friedman test on pooled HV confirms overall differences among the five solvers across all 200 scenarios ($\chi^2 = 669.02$, $p < 10^{-140}$; G-BL and GA-P-BL coincide in 199 of 200 scenarios and never differ to GA-P-BL's disadvantage, so the statistic is conservative toward fewer distinct rankings). G-BL, GA-P-BL, and G-SM each contribute a single point per scenario, so the pooled statistic is driven largely by the within-scenario HV dispersion of the two multi-point NSGA-III solvers rather than by baseline-vs-baseline differences. Pairwise Wilcoxon signed-rank tests use the Bonferroni threshold $\alpha = 0.005$ ($\alpha = 0.05/10$; the four-variant ablation of § 6.5 applies $\alpha = 0.05/12$ per density class); pairs with zero within-pair difference are dropped from the Wilcoxon signed-rank statistic and reported separately; interpretation relies on Cliff's δ given the large sample size. MOEA-2 and MOEA-3 differ on pooled HV with $\delta = -0.56$ ($p < 10^{-5}$), but this contrast blends the sparse-regime divergence with the dense-regime convergence and is inflated by the HV point-count asymmetry (§ 6.4); contrasts involving G-SM are large because it occupies an extreme radiometric-quality regime.

Independently of the multi-objective question, GA-P-BL output is numerically identical to G-BL in 199 of 200 scenarios (never worse by construction), because its never-worse-than-baseline guard (Sec. 4.4) returns the greedy schedule on the single profit objective; the present claims therefore concern the marginal value of adding physical objectives rather than profit-oriented search alone.

7.2. Operational Implications

The scale dependence has direct consequences for mission planning. In sparse regimes ($N \leq 100$), MOEA-2 achieves a +34% geometric-quality improvement at S1 at a 14% coverage cost ($f_1^* = 0.86$ at $N = 20$ relative to G-BL). Whether this trade-off is acceptable depends on the application: for maritime surveillance, the knee-point +34% f_2 gain at S1 corresponds to roughly a 25% reduction in ground pixel area (pixel area scales as $1/f_2$ for the geometric index of Eq. 2), and the coverage cost may be justified; for area-monitoring missions, the greedy baseline remains preferable. In dense regimes ($N \geq 300$), the multi-objective advantage falls below 1% improvement in f_2, while G-SM offers the highest radiometric quality (+16% f_3 at S4) at a 63% coverage cost; this gain comes from the one-line squint-minimization heuristic rather than from multi-objective search.

The agility-envelope sensitivity, previously an open qualification, is tested directly: repeating the sparse $N = 20$ configurations with the window-generation squint cap reduced from 45° to 25° and 15° (50 paired scenarios per cap, identical targets and seeds) shows that the multi-objective edge contracts monotonically with the envelope. Relative to G-BL in the same scenarios, MOEA-3's f_2 advantage shrinks from +28% at the 45° cap to +9% at 25° and +2.3% at 15°, while its coverage cost shrinks from 15% to 4% to $\approx 0\%$ ($f_1^* = 0.85$, 0.96, 1.01); absolute geometry improves for every solver under the tighter cap (G-BL f_2 rises from 0.295 to 0.405 to 0.448), because window filtering removes the low-quality high-squint options. The quality advantage of physics-aware multi-objective scheduling is therefore specific to wide-envelope platforms; on platforms with $|\psi_{\mathrm{sq}}| \lesssim 25°$ agility—or without Doppler-centroid compensation for large squints (§ 3.1)—the greedy baseline attains comparable quality at full coverage even in sparse regimes. The logistic midpoint $N_{50} \approx 50$ (§ 6.3) provides a coarse indicator between the density regimes. Our results suggest a conditional operational guideline: for sparse observation requests ($N \lesssim 100$) on squint-agile platforms (the 45° experimental envelope of §3.1), MOEA-2 may provide geometric quality gains that outweigh the coverage penalty in specific contexts—the knee point trades 14% coverage for +33% f_2, while the best-coverage front point still gives +12.5% f_2 at zero coverage cost, a configuration preferable when coverage cannot be sacrificed; for sparse requests on narrow-envelope platforms, and for dense monitoring campaigns ($N \gtrsim 300$) on any platform, greedy methods (G-SM for radiometric priority, G-BL for coverage priority) attain the tested family's coverage reference and may be preferable. Indeed, the MOEA-2 coverage gap relative to the coverage-only anchor is 0% at $N \geq 300$ (Table 5); since GA-P-BL output coincides with G-BL in 199 of 200 scenarios (and is never worse), no meaningful separate gap exists. The dense-regime greedy recommendation is also latency-favorable: G-BL places each task at its earliest feasible time within the visibility window, whereas quality-steering solvers may delay acquisitions, so the coverage-priority choice aligns with timeliness-sensitive operations.

7.3. Transferability Across Remote-Sensing Domains

The analytic device most likely to transfer beyond SAR is the envelope–null–schedule decomposition of §6.4: separating the geometric correlation over the full feasible envelope from the within-schedule residual isolates where an objective's value is lost to saturation rather than to a mathematical redundancy. The geometric-resolution objective (f_2) carries directly to optical systems, where ground sample distance grows with off-nadir angle (the optical pixel footprint scales as $1/\cos^2 \eta$), and the same density-dependent saturation would appear; the radiometric objective (f_3, NESZ) does not, because optical radiometry depends on illumination and cloud cover rather than on observation geometry alone. The decomposition, rather than any single solver ranking, is the transferable contribution.

7.4. Limitations

Our study is bounded by scope decisions that define directions for future work. The density-dependent third-objective indistinguishability reported above is specific to the single-pass, single-satellite Stripmap setting: it reflects solver convergence to lower-squint geometries that attenuate the envelope-level f_2–f_3 correlation while the residual within-schedule θ-conflict persists (negative at all densities, strongest at sparse density; §6.4, Eq. 6), not a mathematical identity between f_2 and f_3. In multi-pass or multi-satellite settings with greater angular diversity, the θ-conflict would re-emerge and the third objective may become discriminative. Similarly, ScanSAR, Spotlight, and TOPS modes introduce different NESZ and resolution dependencies beyond Stripmap, and the scenario design uses a fixed orbit (693 km circular LEO) without exploring sensitivity to orbital parameters. The hot-start control experiment (§ 6.6) indicates that G-BL initialization materially improves MOEA convergence; the coverage convergence reported here is conditional on this initialization, and full-scale random-init experiments are needed to establish the unconditional behavior. The hot-start dependence implies that the coverage-convergence result reflects the combination of G-BL prior knowledge with MOEA search; whether the MOEA can independently discover high-coverage solutions without greedy seeding remains an open question. Similarly, the NSGA-III crowding-distance explanation for the S1 variant-D reversal (§ 6.5) awaits verification with alternative crowding operators. Two further scope notes bound the operational reading: the energy and memory budgets (C3–C4) are deliberately set far below their caps, so the saturation result holds in the resource-unconstrained regime and may not transfer when energy or downlink capacity is the binding constraint; and the dwell time is fixed at 30 s, removing the dwell-versus-SNR trade-off that an agile scheduler could otherwise exploit. The utility model also optimizes quality and quantity at a fixed timeliness operating point without an explicit latency or urgency objective; the time-critical monitoring motivation of § 1 is therefore addressed only indirectly, through the earliest-feasible placement of the greedy baseline and the short dwell interval, rather than as a distinct scheduling dimension. The reference-point counts of MOEA-2 ($H = 13$) and MOEA-3 ($H = 91$) differ by construction, but a matched control at $p = 4$ ($H = 15$, § 6.4) confirms the sparse-regime f_3 gain is not driven by reference-direction density. Finally, we do not benchmark against learning-based schedulers (e.g., Chen et al. (2025)), which represent an alternative methodological family; a systematic comparison of physics-aware heuristic methods against data-driven approaches is left to future work.

8. Conclusion

We described a physics-aware multi-objective optimization approach for agile SAR satellite scheduling that integrates NESZ radiometric quality and geometric resolution as explicit optimization objectives alongside coverage profit. Five configurations (two greedy heuristics, a single-objective GA control numerically identical to G-BL in 199 of 200 scenarios, and NSGA-III with two and three objectives) were compared on 200 systematically varied scenarios across four density classes. The central finding is a scale-dependent empirical saturation of the marginal benefit of physical objectives in G-BL-seeded MOEA formulations for single-pass, single-satellite Stripmap SAR operations under the tested configuration. At sparse density the three objectives form a genuine trade-off surface (MOEA-2 raises f_2 by +34% (knee-point value; best-coverage front +12.5%) and MOEA-3 raises f_3 by +40% over the greedy anchor at a 14–15% coverage cost), while at high densities the marginal quality gains become negligible and all configurations converge to the G-BL reference. The saturation arises from solver convergence to lower-squint regimes under which the geometric coupling aligns f_2 and f_3 along the shared $\cos\psi_{\mathrm{sq}}$ factor, making the third objective indistinguishable from the two-objective formulation at high densities under the tested search, not at sparse density; this is a convergence effect, and the reported fronts are best-known NSGA-III approximations rather than verified Pareto sets. Coverage convergence toward the G-BL reference is conditional on the hot start; the geometric coupling (visibility-envelope correlation -0.36, restored to the independent-angle null strength of -0.60 in sparse schedules at -0.63, and collapsing to -0.2 to -0.15 at high density within schedules) is the anchor-independent result (§ 6.4). GA-P-BL output coincides with the G-BL reference in 199 of 200 scenarios, because its never-worse-than-baseline guard triggers on the single profit objective in every scenario but one, so this result delimits the benefit of adding physical objectives. This saturation regime applies to the evaluated target-density range ($N = 20$–500) and G-BL-seeded NSGA-III search. For mission planning, the operational implication is conditional: multi-objective optimization provides measurable benefit only when observation requests are sparse; at dense monitoring cadences, greedy methods attain the tested family's coverage reference. These findings establish an empirical, conditional saturation regime for what physics-aware multi-objective scheduling can contribute to single-satellite SAR operations, and provide a baseline against which multi-satellite, multi-pass, and multi-mode extensions can be measured.

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author used AI-assisted tools (Claude, Codex) for manuscript preparation and revision assistance only. The author reviewed and edited all AI-assisted output and takes full responsibility for the content of this article.

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CRediT Authorship Contribution Statement

Liu Shuhao: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Resources, Data Curation, Writing – Original Draft, Writing – Review & Editing, Visualization.

Data Availability Statement

Solver code (NSGA-III via `pymoo` 0.6.1.6), scenario generators, fixed seeds and configurations, processed JSON/CSV results, analysis scripts, and figures are available from the corresponding author. The companion repository is publicly available at `https://github.com/liushuhao/sar-quality-aware-scheduling`. Generated binary scenario files and per-run pickle objects are not tracked; they can be regenerated using the supplied scripts.

Declaration of Competing Interest

The author declares that he has no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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